

Are galaxies in compact groups special?

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ABSTRACT

Context. Compact groups combine high projected densities with low velocity dispersions, but many are embedded in or projected onto larger structures. Their evolutionary significance depends on whether their galaxies remain distinct from regular-group analogues.

Aims. We test whether CG₄ galaxies remain special after removing split systems and comparing them with ordinary four-member groups, luminous regular-group cores, and projected compact cores. We ask which observable carries any residual signal: morphology, star formation, colour, AGN-like activity, group structure, phase-space position, or local projected tidal density.

Methods. We use the CG₄ sample inherited from Tricottet et al., leaving 62 compact groups after removing split systems likely to be projected or multi-component configurations, and compare it with RG₄, Control_{4B}, and Control_{4C}. We combine Galaxy Zoo morphologies, SDSS masses and specific star-formation rates, colours, emission-line diagnostics, adjusted binomial models, and matched controls.

Results. The robust residual signal is satellite morphology. In adjusted satellite models, CG₄ membership raises the odds of an elliptical/smooth classification by a factor 1.84 and lowers the odds of a spiral/features classification by a factor 0.54. The BGG morphology coefficient is not significant after the same adjustments. The star-formation signal is weaker: CG₄ satellites have a lower raw star-forming fraction, especially relative to RG₄, but this contrast is not comparably robust after adjustment and matching. Star-forming galaxies show no robust compact-group offset from the star-forming main sequence. Colours, AGN-like fractions, magnitude gaps, and limited H α diagnostics do not support a simple instantaneous-quenching, fossil-group, or AGN-driven explanation. A projected pairwise tidal-index diagnostic suggests that local projected density accounts for part of the morphology signal.

Conclusions. Compact groups host a more morphologically evolved population, consistent with accumulated transformation in dense group cores and/or preprocessing, favouring a cumulative structural signal over a uniquely isolated compact-group quenching mode.

Key words. galaxies: groups: general – galaxies: evolution – galaxies: interactions – galaxies: star formation – galaxies: elliptical and lenticular, cD – methods: statistical

1. Introduction

Compact groups of galaxies are useful laboratories for environmental evolution because they combine high projected galaxy densities with relatively low velocity dispersions (Hickson 1982; Hickson et al. 1992). In such configurations, repeated tidal encounters, dynamical friction, mergers, gas stripping or consumption, and disk heating can plausibly accelerate the transformation of galaxies compared with more diffuse environments (Barnes 1985, 1989; Verdes-Montenegro et al. 2001). This expectation is closely connected to the broader morphology-density and star-formation-environment relation, in which galaxy structure and star-formation activity vary systematically with local density (Dressler 1980).

The interpretation of compact groups is nevertheless ambiguous. Projected selection can include chance alignments, and many Hickson-like systems need not be isolated bound entities. Simulations and redshift-space anal-

yses have long shown that apparent compact groups may be transient alignments, dense substructures, or cores embedded in looser groups and richer haloes (Mamon 1986; Díaz-Giménez & Mamon 2010; Díaz-Giménez et al. 2012; Zheng & Shen 2021). The relevant observational question is therefore not only whether compact-group galaxies differ from field galaxies, but whether they remain different from galaxies occupying regular-group analogues selected with comparable luminosity, redshift, and membership constraints.

Recent structural work strengthens this motivation. Using S-PLUS imaging and single-Sérsic measurements, Montaguth et al. (2023, 2025, 2026b,a) identified transition galaxies and trends in the R_e -Sérsic-index plane that point to structural transformation in compact groups, with stronger signatures in non-isolated compact groups and host structures. These studies suggest that compact groups can mark sites of cumulative structural processing. They do not, however, by themselves answer whether compact-group

galaxies remain distinct after comparison with regular-group cores selected from the same parent catalogues.

This structural perspective complements earlier compact-group galaxy-population work. Studies of Hickson-like compact groups have reported enhanced early-type fractions, suppressed or transformed star formation, mid-infrared colour bimodality, and gas processing or HI deficiency in at least part of the population (e.g. Johnson et al. 2007; Walker et al. 2010; Tzanavaris et al. 2010; Bitsakis et al. 2010, 2011; Coenda et al. 2012; Plauchu-Frayn et al. 2012; Verdes-Montenegro et al. 2001). The central question for the present work is therefore not whether compact groups differ from the field, but whether any such population difference remains when compact groups are compared with regular-group cores and close projected companions selected from the same parent catalogues.

Our previous sample-construction paper (Tricottet et al. 2025) showed that many compact groups are not isolated systems, but appear as dense cores or substructures of richer environments. This result motivates the stricter comparison made here. We use the same compact-group and regular-group sample framework to ask whether CG galaxies are still special after controlling for stellar mass, redshift, galaxy rank, and available group-scale properties. We then ask which observable carries the signal: star-formation class, Galaxy Zoo morphology, optical colour, AGN-like activity, fossil-like magnitude gaps, projected phase-space position, or local projected tidal density.

The paper is organized as follows. Section 2 describes the samples, the sSFR and morphology classifications, and the statistical hierarchy. Sections 3 and 4 present baseline population differences and group-scale diagnostics. Section 5 tests which signals survive adjusted and matched comparisons and examines secondary diagnostics. Section 6 discusses the physical interpretation of the morphology signal, and Section 7 summarizes the conclusions.

2. Data and classifications

2.1. Samples

We use the compact-group and control samples constructed in our previous study Tricottet et al. (2025). We briefly recall their construction here. Galaxy properties are taken from the SDSS DR12 catalogue of Tempel et al. (2017), while regular groups are drawn from the SDSS DR7 group catalogue of Lim et al. (2017); the two catalogues were matched through SDSS object identifiers to define the Lim–Tempel working catalogue. Compact groups are selected from the modified Hickson compact-group catalogue of Díaz-Giménez et al. (2018). Starting from the reliable HMCs with exactly four members, we retain only systems in which all galaxies are present in the Lim–Tempel catalogue and impose the volume- and luminosity-complete limits $0.005 < z_{\text{BGG}} < 0.0452$ and $M_{r,\text{BGG}} \leq -21.81$, with all 4 galaxies lying within 3 magnitudes of the brightest group galaxy. This defines the parent CG₄ sample of 78 compact groups. The regular-group comparison samples are built from Lim–Tempel groups satisfying the same redshift and BGG-luminosity limits, with at least 4 members and at least 3 satellites within 3 magnitudes of the BGG. From this parent control catalogue, we define RG₄ groups as reg-

ular groups with exactly four members, excluding systems identical to a CG₄, and we additionally use the Control_{4B} and Control_{4C} samples, formed respectively from the 4 brightest galaxies and from the BGG plus its 3 closest projected companions in each parent regular group, after removing groups that share galaxies with CG₄s. In contrast to our previous study, we remove CGs classified as *split* following Zheng & Shen (2021). These systems are likely chance alignments or multi-component projections, and retaining them can inflate projected compactness and dynamical quantities such as velocity-dispersion and crossing-time proxies (Tricottet et al. 2025; Zheng & Shen 2021). This reduces the compact-group sample from 78 to 62 groups. We also remove the systems associated with Lim group 3688 from our Control_{4B} and Control_{4C} samples. This group contains 6 galaxies with redshifts between 0.032 and 0.034, and 1 with a redshift of 0.048. This group appears to be spurious, and the outlying galaxy, although not the BGG, is selected in both our original Control_{4B} and Control_{4C} samples, significantly affecting the group velocity dispersion (1979 and 1981 km s⁻¹, respectively).

The three regular-group controls serve different physical purposes. RG₄ contains true regular groups with exactly four selected members and provides an equal-membership, comparatively low-density anchor. Control_{4B} contains the four brightest galaxies in richer regular groups and tests whether the CG₄ population resembles the luminous core of an ordinary group. Control_{4C} contains the BGG plus the three closest projected companions and is the most compact-core-like ordinary-group analogue. The CG₄–Control_{4C} comparison is therefore the strictest test of whether compact groups differ from ordinary projected group cores, while RG₄ is useful for bracketing the contrast with genuine four-member regular groups.

We use SDSS DR 16 to retrieve stellar masses, star formation rates (SFRs), and Galaxy Zoo morphologies (Lintott et al. 2011). We specifically extract the columns `sfr_tot_p50`, `specsfr_tot_p50`, and `lgm_tot_p50` from the `galSpecExtra` table, and `p_el_debiased` and `p_cs_debiased` from the `zooSpec` table. The `galSpecExtra` stellar masses and SFRs are the SDSS/MPA-JHU value-added products, based on the stellar-population and emission-line methodology of Kauffmann et al. (2003) and Brinchmann et al. (2004); we use them on their native Kroupa-IMF scale. Following the parent sample construction, projected physical distances recomputed in this work use the flat Planck cosmology ($\Omega_m = 0.308$, $h = 0.678$) adopted by Tricottet et al. (2025) from Planck Collaboration et al. (2016), while published catalogue group quantities are used as tabulated. Any small residual h -convention mismatch is negligible for the differential comparisons emphasized here.

The “SDSS selection” row used in the tables and figures denotes the parent SDSS comparison set extracted with the same redshift and apparent-magnitude limits before applying the compact- or regular-group selections. It is used mainly to calibrate the broad sSFR classification and to provide a reference distribution, not as the primary control sample.

Because the diagnostics below require different catalogue fields, the relevant parent N differs slightly between analyses. For example, the sSFR and morphology tables use the galaxies with the corresponding SDSS value-added quantities, the colour analysis requires complete matched

photometry, the BGG-distance analysis requires projected-distance information for satellites, and the matched comparison further requires complete propensity-score covariates and common support. We therefore report the relevant N for each diagnostic; the selection audit in Sect. 5.9 summarizes the main bookkeeping.

2.2. *sSFR* classification

To build a general star formation classification that could be uniformly applied to all our samples, we extracted galaxies from the SDSS DR 16 using the query shown in Listing 1.

```

1  SELECT
2      s.specObjID,
3      s.z,
4      p.petroMag_r - p.extinction_r AS
5          r_obs,
6      p.petroMag_u - p.extinction_u AS
7          u_obs,
8      p.petroMag_g - p.extinction_g AS
9          g_obs,
10     p.petroMag_i - p.extinction_i AS
11     i_obs,
12     p.petroMag_z - p.extinction_z AS
13     z_obs,
14     p.objID,
15     g.sfr_tot_p50, g.specsfr_tot_p50, g.
16     lgm_tot_p50,
17     l.h_alpha_eqw, l.h_beta_eqw, l.
18     oiii_5007_eqw, l.nii_6584_eqw,
19     l.h_alpha_flux, l.h_beta_flux, l.
20     oiii_5007_flux, l.nii_6584_flux,
21     z.p_el_debiased AS p_E,
22     z.p_cs_debiased AS p_S
23 FROM SpecObj AS s
24 JOIN PhotoObj AS p ON s.bestObjID = p.
    objID
25 JOIN galSpecExtra AS g ON s.specObjID
    = g.specObjID
26 JOIN galSpecLine AS l ON s.specObjID
    = l.specObjID
27 JOIN zooSpec AS z ON s.specObjID = z.
    specObjID
28 WHERE s.z BETWEEN 0.005 AND 0.0452
29 AND (p.petroMag_r - p.extinction_r <=
30     17.77)
31 AND s.class = 'GALAXY'
32 AND g.lgm_tot_p50 > -1000
    
```

Listing 1. Query used for selecting spectral & photometric data from the SDSS

We identify BPT-classifiable galaxies by requiring measured values of both emission-line ratios

$$\log_{10}([\text{NII}]\lambda 6584/\text{H}\alpha) \text{ and } \log_{10}([\text{OIII}]\lambda 5007/\text{H}\beta).$$

Among these galaxies, we flag broad AGN-like systems if they satisfy either

$$\log_{10} \frac{[\text{NII}]}{\text{H}\alpha} > 0,$$

or if they lie above the empirical demarcation of Kauffmann et al. (2003):

$$\log_{10} \frac{[\text{OIII}]}{\text{H}\beta} > \frac{0.61}{\log_{10}([\text{NII}]/\text{H}\alpha) - 0.05} + 1.3,$$

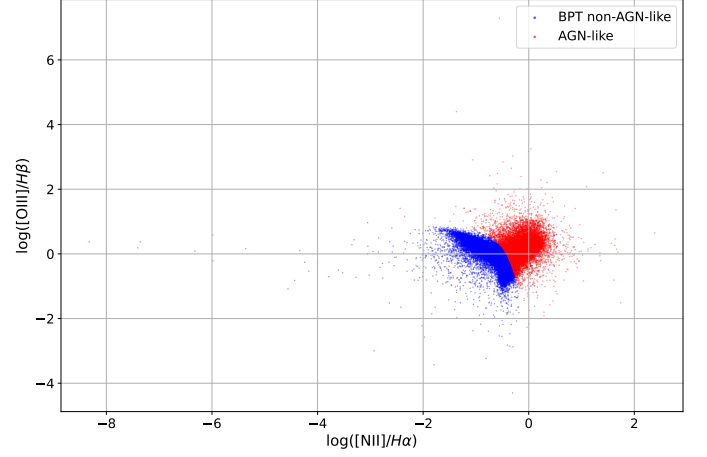


Fig. 1. BPT diagram for galaxies with measured diagnostic line ratios. The AGN-like region is excluded from the *sSFR* calibration sample; galaxies without the required ratios are BPT-unclassified rather than classified as non-AGN by this diagram.

in which case they are excluded from the GMM calibration sample. Galaxies with missing diagnostic line ratios are BPT-unclassified, not proven non-AGN. They are retained for the broad *sSFR* status assignment when valid SDSS stellar mass and *sSFR* estimates are available, while the emission-line/AGN-like analysis below treats BPT-unclassified objects separately. The BPT-classifiable subset is shown in Fig. 1.

Galaxies with `specsfr_tot_p50` equal to -9999 in the SDSS are classified as quenched. In figures that require a continuous ordinate, these objects are shown at the arbitrary plotting floor $10^{-14.0} \text{ yr}^{-1}$; this value is used only for display and is not interpreted as a measured *sSFR*.

The remaining galaxies are separated into star-forming and passive populations with a two-component Gaussian Mixture Model (GMM) in the $\log M_*$ – $\log s\text{SFR}$ plane. The model is calibrated on galaxies not flagged as AGN-like when the required BPT ratios are available, while BPT-unclassified galaxies with valid SDSS stellar mass and *sSFR* estimates can still receive a broad *sSFR* class. The GMM provides a data-driven division between the high-*sSFR* sequence and the low-*sSFR* cloud, in the spirit of previous multi-Gaussian decompositions of the $\text{SFR}-M_*$ plane (McLachlan & Peel 2000; Wuyts et al. 2011; Hahn et al. 2019). We use the equal-posterior boundary between the two components as the passive/star-forming separator. The exact optimizer settings are not used in the interpretation; the key results are checked below against a simpler fixed-*sSFR* threshold.

2.3. Raw *sSFR* fractions

The *sSFR* status for each sample is shown in Table 1. In the raw all-galaxy comparison, Fisher’s exact test gives star-forming-fraction p -values of $p = 0.648$ versus the $\text{Control}_{4\text{B}}$ sample, $p = 0.25$ versus $\text{Control}_{4\text{C}}$, and $p = 0.018$ versus RG_4 . We treat these raw tests as descriptive because they do not adjust for stellar mass, rank, or group-scale differences.

The same comparison restricted to BGGs and satellites is also shown in Table 1. For BGGs, the p -values obtained with two-sided Fisher exact test are $p = 0.578$ against

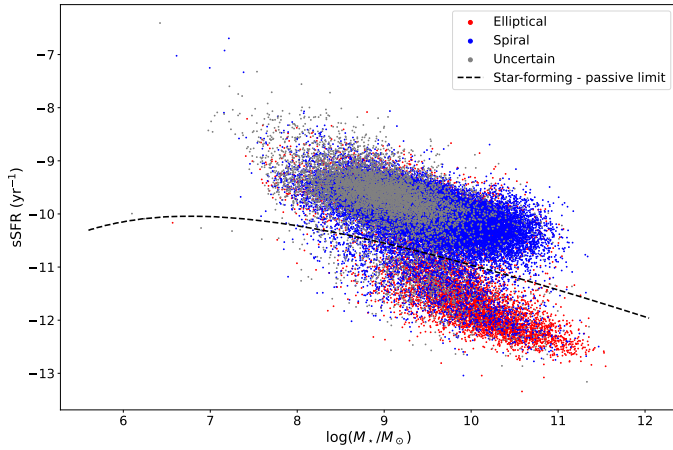


Fig. 2. sSFR vs mass, star-forming/passive limit and Zoo morphologies for the SDSS control sample, excluding quenched galaxies.

Control_{4B}, $p = 0.587$ against Control_{4C} and $p = 0.37$ against RG₄. For satellites, the corresponding p-values are $p = 0.303$ against Control_{4B}, $p = 0.033$ against Control_{4C} and $p = 2.3 \times 10^{-4}$ against RG₄. The satellite deficit is clearest in the raw comparison with RG₄ and is revisited below with adjusted and matched analyses.

2.4. Morphology classification

Morphologies are determined via the Galaxy Zoo citizen-science decision tree, in which each galaxy image receives multiple independent volunteer classifications that are aggregated into debiased vote fractions (Lintott et al. 2011). Those fractions are provided in the zooSpec table of the SDSS database through the `p_el_debiased` and `p_cs_debiased` columns. We classify a galaxy as “elliptical/smooth” if `p_el_debiased` is greater than 0.5, and as “spiral/features” if `p_cs_debiased` is greater than 0.5. All other cases are classified as “uncertain”. These are vote-fraction classes, not a full structural decomposition. In particular, the catalogue used here does not provide a clean E/S0 separation, so references below to elliptical or early-type galaxies should be read as Galaxy Zoo smooth/early-type proxies rather than as a bulge-disc structural classification.

2.5. Statistical strategy

The analyses are organized hierarchically. Raw tables and exact tests provide descriptive baseline comparisons. Fisher or Barnard exact tests are used for two-sample categorical comparisons depending on the table structure, while rank-sum or bootstrap comparisons are used for distributional and median differences. Spearman and Pearson correlations are used only as diagnostics of monotonic or linear trends; when the coefficients are small, we interpret them as weak even if the p-values are formally small. Holm or Benjamini-Hochberg corrections are applied where multiple related tests are performed. The primary robustness layer consists of mass-, rank-, and environment-adjusted binomial models with group-clustered standard errors, followed by one-to-one matched control comparisons. Colours, AGN-like fractions, H α equivalent widths, magnitude gaps, crossing-time

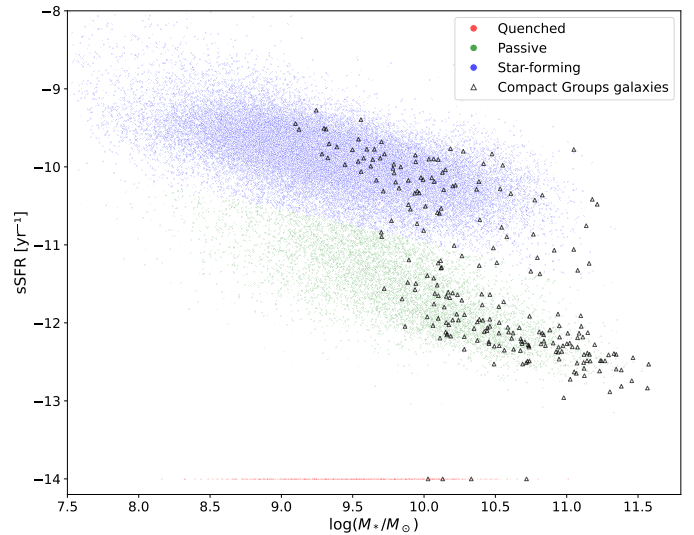


Fig. 3. sSFR and morphologies of galaxies in CG₄ and our SDSS selection. Quenched galaxies are represented at an arbitrary low value of $10^{-14.0} \text{ yr}^{-1}$.

trends, projected phase space, and the pairwise tidal index are secondary diagnostics. Their weak or null results should be interpreted as non-detections of robust offsets in the present data and under the adopted measurements, not as evidence that such effects are physically absent. The main conclusions below rely on signals that survive the adjusted and matched analyses, not on isolated exploratory tests.

3. Baseline population differences

3.1. Raw morphology fractions

The morphologies obtained for each sample are shown in Table 2. Barnard’s two-sided exact tests were used to compare the fraction of each morphology between CG₄ and each control sample. The resulting p-values are $p = 1.4 \times 10^{-3}$ for ellipticals and $p = 1.2 \times 10^{-3}$ for spirals versus the Control_{4B} sample, $p = 3.3 \times 10^{-3}$ and $p = 0.034$ versus Control_{4C} and $p < 10^{-6}$ and $p < 10^{-6}$ versus RG₄. The galaxies in CG₄ systems, which we identified as cores of richer groups in Tricottet et al. (2025), show a robust raw excess of elliptical/smooth classifications and a deficit of spiral/features classifications. Figure 3 displays sSFR versus stellar mass for galaxies in the SDSS selection and in CG₄.

3.2. Morphology versus sSFR

Among star-forming CG₄ galaxies, 58 are classified as spiral (84.1% after removing galaxies with uncertain morphology) and 11 are classified as elliptical (15.9% after removing galaxies with uncertain morphology). The corresponding numbers are 809 spiral galaxies (88.4%) and 106 elliptical galaxies (11.6%) for Control_{4B}, 851 spiral galaxies (84.3%) and 159 elliptical galaxies (15.7%) for Control_{4C}, and 91 spiral galaxies (91%) and 9 elliptical galaxies (9%) for RG₄. The probabilities that these fractions are drawn from the same distribution, estimated with Barnard two-sided exact test, are $p = 0.287$ against Control_{4B}, $p = 0.969$ against Control_{4C}, and $p = 0.186$ against RG₄. We therefore

Sample		Quenched	Passive	Star-forming
CG ₄		18 (7.3 %)	146 (58.9 %)	84 (33.9 %)
	<i>BGG</i>	6 (9.7 %)	45 (72.6 %)	11 (17.7 %)
	<i>Satellites</i>	12 (6.5 %)	101 (54.3 %)	73 (39.2 %)
Control _{4B}		181 (6.5 %)	1596 (57.1 %)	1019 (36.4 %)
	<i>BGG</i>	58 (8.3 %)	533 (76.4 %)	107 (15.3 %)
	<i>Satellites</i>	122 (5.8 %)	1061 (50.7 %)	911 (43.5 %)
Control _{4C}		191 (6.3 %)	1619 (53.8 %)	1198 (39.8 %)
	<i>BGG</i>	65 (8.7 %)	568 (75.6 %)	118 (15.7 %)
	<i>Satellites</i>	125 (5.5 %)	1050 (46.6 %)	1078 (47.8 %)
RG ₄		5 (2.2 %)	101 (45.1 %)	118 (52.7 %)
	<i>BGG</i>	3 (5.4 %)	38 (67.9 %)	15 (26.8 %)
	<i>Satellites</i>	2 (1.2 %)	63 (37.5 %)	103 (61.3 %)
<i>SDSS selection</i>		634 (1.2 %)	7705 (14.7 %)	44192 (84.1 %)

Table 1. Number of galaxies in each sSFR status for each sample. Each diagnostic table reports the availability-limited sample relevant to that measurement, so parent totals can differ slightly between tables.

Sample	Elliptical/smooth	Spiral/features	Uncertain
CG ₄	124 (50.0 %)	86 (34.7 %)	38 (15.3 %)
Control _{4B}	1106 (39.6 %)	1266 (45.3 %)	420 (15.0 %)
Control _{4C}	1215 (40.4 %)	1249 (41.6 %)	540 (18.0 %)
RG ₄	60 (26.8 %)	129 (57.6 %)	35 (15.6 %)
<i>SDSS selection</i>	10091 (19.2 %)	34715 (66.1 %)	7725 (14.7 %)

Table 2. Number of galaxies in each Galaxy Zoo vote-fraction morphology proxy for each sample. As in Table 1, totals reflect the availability-limited sample for this diagnostic.

find neither a significant excess nor a deficit of star-forming elliptical galaxies in CG₄ relative to any control sample.

When restricting the comparison to BGGs, we find 36 elliptical BGGs (58.1% of the BGGs) and 17 spiral BGGs (27.4%) in CG₄. The corresponding numbers are 405 elliptical BGGs (58.0%) and 195 spiral BGGs (27.9%) in Control_{4B}, 424 elliptical BGGs (56.5%) and 216 spiral BGGs (28.8%) in Control_{4C}, and 21 elliptical BGGs (37.5%) and 26 spiral BGGs (46.4%) in RG₄. The p-values obtained with two-sided Fisher exact test are 1.00 against Control_{4B}, 0.88 against Control_{4C} and 0.03 against RG₄. Thus, BGG morphologies differ between CG₄ and RG₄: CG₄ BGGs have a higher elliptical fraction than RG₄ BGGs.

3.3. Population medians by sSFR class

Tables 3, 4, and 5 summarize the typical stellar masses, absolute magnitudes, and sSFRs of passive and star-forming galaxies in the different samples, including splits by the sSFR class of the BGG. These medians are descriptive rather than causal: they show that star-formation class is entangled with stellar mass, luminosity, and group role. This motivates the adjusted models in Sect. 5, where stellar mass, rank, and group-scale quantities are treated explicitly.

3.4. Main sequence

We model the star-forming main sequence as an empirical function of stellar mass using polynomial orders 1–4; the polynomial-order comparison is shown in Appendix Fig. A.5. We adopt the quadratic fit as a flexible but still

smooth reference relation for residual measurements. The qualitative conclusions below do not rest on interpreting this polynomial as a physical model of the main sequence. We then compute the residual of each star-forming galaxy with respect to this reference sequence and show the resulting distributions in Fig. 4. Positive residuals correspond to higher sSFR at fixed stellar mass relative to the fitted sequence.

The median main-sequence residual of CG₄ galaxies is:

- marginally different from that of Control_{4B}, with a median offset of $\Delta = 0.056$ ($0.027 \leq \Delta \leq 0.098$, 16th–84th percentile bootstrap interval), with the associated rank-sum comparison giving $p = 0.058$.
- significantly different from that of Control_{4C}, with a median offset of $\Delta = 0.082$ ($0.062 \leq \Delta \leq 0.123$, 16th–84th percentile bootstrap interval), with the associated rank-sum comparison giving $p = 0.011$.
- not significantly different from that of RG₄, with a median offset of $\Delta = 0.046$ ($-0.037 \leq \Delta \leq 0.098$, 16th–84th percentile bootstrap interval), with the associated rank-sum comparison giving $p = 0.532$.

The direct rank-sum comparisons of these residuals are used only as a consistency check and do not change the bootstrap-based interpretation above.

Within CG₄, the star-forming main-sequence offsets of the Embedded, Isolated and Predominant classes remain statistically consistent with one another.

Sample	$\log_{10}(\text{sSFR} [\text{yr}^{-1}])$		M_r		$\log(M_\star/M_\odot)$	
	Passive	Star-forming	Passive	Star-forming	Passive	Star-forming
CG4	-12.08 ± 0.03	-10.19 ± 0.07	-21.16 ± 0.14	-20.4 ± 0.14	10.64 ± 0.07	10.05 ± 0.06
Control4B	-12.1 ± 0.01	-10.31 ± 0.02	-21.52 ± 0.05	-20.75 ± 0.04	10.79 ± 0.01	10.24 ± 0.02
Control4C	-11.92 ± 0.02	-10.16 ± 0.02	-20.81 ± 0.07	-19.93 ± 0.04	10.47 ± 0.03	9.82 ± 0.03
RG4	-11.91 ± 0.07	-10.16 ± 0.07	-21.11 ± 0.25	-20.32 ± 0.12	10.62 ± 0.08	9.97 ± 0.09

Table 3. Median values of galaxy properties for passive and star-forming populations in each sample. Uncertainties are estimated by bootstrapping (10 000 iterations).

Sample	$\log_{10}(\text{sSFR} [\text{yr}^{-1}])$		M_r		$\log(M_\star/M_\odot)$	
	Passive	Star-forming	Passive	Star-forming	Passive	Star-forming
CG4	-11.53 ± 0.09	-10.85 ± 0.29	-20.93 ± 0.11	-20.77 ± 0.32	10.46 ± 0.06	10.32 ± 0.12
Control4B	-11.53 ± 0.03	-10.86 ± 0.06	-21.25 ± 0.03	-21.12 ± 0.07	10.61 ± 0.02	10.46 ± 0.04
Control4C	-11.3 ± 0.03	-10.62 ± 0.06	-20.42 ± 0.05	-20.55 ± 0.08	10.22 ± 0.02	10.15 ± 0.06
RG4	-11.14 ± 0.19	-10.58 ± 0.16	-20.69 ± 0.14	-20.74 ± 0.28	10.32 ± 0.07	10.19 ± 0.12

Table 4. Median values of various quantities for galaxies according to the sSFR class of their BGG (BGG included). Uncertainties are estimated by bootstrapping (10 000 iterations).

Sample	$\log_{10}(\text{sSFR} [\text{yr}^{-1}])$		M_r		$\log(M_\star/M_\odot)$	
	Passive	Star-forming	Passive	Star-forming	Passive	Star-forming
CG4	-11.23 ± 0.16	-10.92 ± 0.48	-20.45 ± 0.13	-20.34 ± 0.18	10.22 ± 0.05	10.09 ± 0.07
Control4B	-11.26 ± 0.06	-10.91 ± 0.13	-20.9 ± 0.03	-20.75 ± 0.05	10.43 ± 0.02	10.3 ± 0.03
Control4C	-10.96 ± 0.04	-10.6 ± 0.07	-19.8 ± 0.04	-20.02 ± 0.10	9.91 ± 0.02	9.89 ± 0.07
RG4	-10.82 ± 0.19	-10.61 ± 0.22	-20.23 ± 0.16	-20.3 ± 0.13	10.09 ± 0.08	10 ± 0.13

Table 5. Median values of various quantities for galaxies according to the sSFR class of their BGG (satellites only). Uncertainties are estimated by bootstrapping (10 000 iterations).

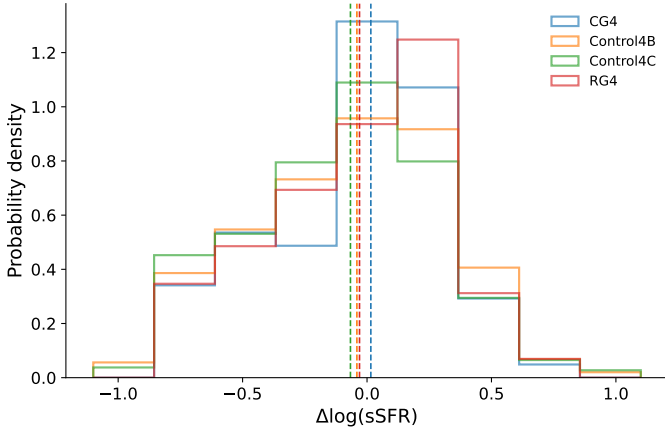


Fig. 4. Main-sequence residual distributions for the star-forming galaxies in each sample.

4. Environmental and group-scale diagnostics

4.1. Distance to the BGG

We also compare the sSFR of satellite galaxies with their distance to the BGG, expressed both in projected kpc and in units of $\langle R_{ij} \rangle$.

Significant monotonic trends are detected in the following cases.

In CG₄, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_s = 0.24$ for 177 satellites

($p = 1.2 \times 10^{-3}$) and Pearson coefficient $r = 0.21$ ($p = 4.1 \times 10^{-3}$).

In Control_{4B}, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_s = 0.08$ for 1991 satellites ($p = 3.6 \times 10^{-4}$) and Pearson coefficient $r = 0.08$ ($p = 2.3 \times 10^{-4}$).

In Control_{4C}, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_s = 0.25$ for 2143 satellites ($p < 10^{-6}$) and Pearson coefficient $r = 0.20$ ($p < 10^{-6}$).

In Control_{4C}, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_s = 0.10$ for 2143 satellites ($p = 4.6 \times 10^{-6}$) and Pearson coefficient $r = 0.09$ ($p = 1.8 \times 10^{-5}$).

The supporting distance plots are shown in Appendix Figs. A.6 and A.7.

4.2. Dominance

We split groups into dominated systems, where the BGG contributes more than 60% of the group luminosity, and non-dominated systems. This split does not produce a strong compact-group signal by itself: CG₄ shows no significant dominated/non-dominated difference after the Benjamini–Hochberg false-discovery-rate (FDR) correction. Here p_{adj} denotes the FDR-adjusted p-value, and Cliff’s δ is a rank-based effect size ranging from -1 to 1.

Across the control and reference samples, the clearest binary separation is instead in group luminosity. Dominated systems have lower L_{group} than their non-dominated coun-

terparts: Control_{4B} gives 9.1×10^{10} versus $1.28 \times 10^{11} L_{\odot}$ ($p_{\text{adj}} < 10^{-6}$, $\delta = -0.55$); Control_{4C} gives 8.49×10^{10} versus $9.54 \times 10^{10} L_{\odot}$ ($p_{\text{adj}} < 10^{-6}$, $\delta = -0.25$); and RG₄ gives 7.09×10^{10} versus $9.82 \times 10^{10} L_{\odot}$ ($p_{\text{adj}} = 7.8 \times 10^{-5}$, $\delta = -0.59$). The negative δ values mean that dominated groups tend to occupy the lower-luminosity side of each comparison.

The spiral-fraction correlations show a broader environmental pattern rather than a clean dominated/non-dominated dichotomy. The most coherent set appears in Control_{4C}, where S_{frac} increases with $\langle R_{ij} \rangle$ in both dominated and non-dominated groups ($\rho_S = 0.38$, $p < 10^{-6}$ and $\rho_S = 0.35$, $p < 10^{-6}$), while weaker negative trends are found with σ_v and L_{group} . Control_{4B} contributes only weaker non-dominated-group correlations. In CG₄, the significant trends occur in small subsets, with S_{frac} decreasing with L_{group} among dominated groups ($\rho_S = -0.50$, 36 groups) and with $\Delta V_{\text{BGG}}/\sigma_v$ among non-dominated groups ($\rho_S = -0.53$, 26 groups); we therefore treat these as supporting diagnostics rather than the main dominance result. The full distributional and correlation diagnostics are shown in Appendix Figs. A.8 and A.9.

4.3. Morphology and domination

When we further split the samples by domination and by the morphology of the BGG, significant effects only appear in

Control_{4B} dominated groups, where the median satellite spiral fraction is 1.00 for spiral BGGs and 0.67 for elliptical BGGs. The dominated-group satellite counts imply a BGG-morphology odds-ratio point estimate of 1.842; the grouped-binomial summary gives $p = 0.01$, while the group-clustered satellite-level standard-error treatment gives $p = 0.02$;

Control_{4C} dominated groups, where the median satellite spiral fraction is 0.67 for spiral BGGs and 0.50 for elliptical BGGs. The dominated-group satellite counts imply a BGG-morphology odds-ratio point estimate of 1.840; the grouped-binomial summary gives $p = 2.1 \times 10^{-5}$, while the group-clustered satellite-level standard-error treatment gives $p = 2.7 \times 10^{-5}$.

The point estimate is the same because both summaries are based on the same dominated-group satellite counts; the two p-values differ because the second fit uses group-clustered standard errors.

4.4. Crossing time

The crossing-time diagnostic mainly separates systems by dynamical scale rather than by luminosity alone. Shorter t_{cross} systems have larger virial mass-to-light ratios and virial masses: the correlations with $\log(M_{\text{VT}}/L_r)$ and $\log(M_{\text{VT}})$ are $\rho_S = -0.39$ ($p < 10^{-6}$) and $\rho_S = -0.33$ ($p < 10^{-6}$), respectively, over 816 groups. By contrast, the relation with total group luminosity is formally significant but very weak ($\rho_S = 0.08$, $p = 0.03$). We therefore read t_{cross} as a dynamical compactness indicator tied primarily to virial mass and mass-to-light ratio, with little independent leverage from L_{group} . Appendix Figs. A.10–A.12 show the corresponding projected and three-dimensional trends.

4.5. Optical colours

We derive the $(u - r)$, $(u - g)$, $(g - r)$ and $(r - i)$ colours from the matched SDSS photometry. The retained matches include 110 of 248 galaxies in CG₄, 931 of 2792 galaxies in Control_{4B}, 1478 of 3004 galaxies in Control_{4C}, 102 of 224 galaxies in RG₄. Stellar mass is a confounding variable in colour comparisons (Alpaslan et al. 2015); we therefore model each colour jointly with $\log(M_{\star}/M_{\odot})$ rather than comparing unadjusted colour distributions.

The colour-matched subset is not representative of the full galaxy catalogues: matched galaxies are systematically less massive, more actively star-forming, and more frequently satellites than unmatched galaxies. The fractions with all four colours are 44.4% in CG₄, 33.3% in Control_{4B}, 49.2% in Control_{4C}, 45.5% in RG₄. The colour analysis is therefore useful as a secondary diagnostic, but it cannot carry the main interpretation by itself.

The raw catalogue colour-mass fits indicate that CG₄ does not simply follow the same colour-mass relation as every comparison sample: common slopes are rejected in 4 of the four colours, with the significant directional slope differences concentrated in the bluer colours and dependent on the adopted comparison catalogue. In the satellite-only comparison, the fitted offsets and slopes likewise depend on the control definition. These tests motivate the robustness analysis rather than establishing a stand-alone compact-group colour effect.

We therefore refit the satellite colours with cluster-robust standard errors, using catalogue-qualified group identifiers. In the pooled comparison adjusted only for stellar mass and redshift, none of the four CG₄ coefficients is significantly negative after Holm correction. After additionally controlling for morphology and sSFR class, negative coefficients appear in $(u - r)$ ($\Delta = -0.127$ mag, Holm-adjusted $p = 0.018$) and $(u - g)$ ($\Delta = -0.101$ mag, Holm-adjusted $p = 0.027$). However, these adjusted coefficients are control-sample dependent, and they do not persist within the sSFR or morphology splits. The data therefore do not localize the colour effect to the star-forming spiral population expected for a simple interaction-triggered star-formation scenario.

Residual colour trends with projected BGG-centric distance are consistent with stronger processing near the group centre and a less processed outer population, but these trends use the same biased colour subset and are treated as exploratory. The group-level BGG-satellite colour concordance is similarly secondary: redder BGGs tend to have redder mean satellite populations in the combined group sample, but the CG₄ slope is not a robustly distinct population-level signature. Detailed colour-mass, satellite, radial, BGG-satellite, and domination diagnostics are shown in Appendix Figs. A.13–A.17. Figure 5 summarizes the model and control-sample dependence that motivates our interpretation: the present data do not establish a robust global blue or red offset for CG₄ satellites.

5. Robustness: what drives the compact-group differences?

The previous sections establish that CG₄ galaxies differ from the control catalogues in their population mix, especially in morphology and in the star-forming fraction of satellites. We now ask whether these differences survive after controlling for stellar mass, rank and available group-

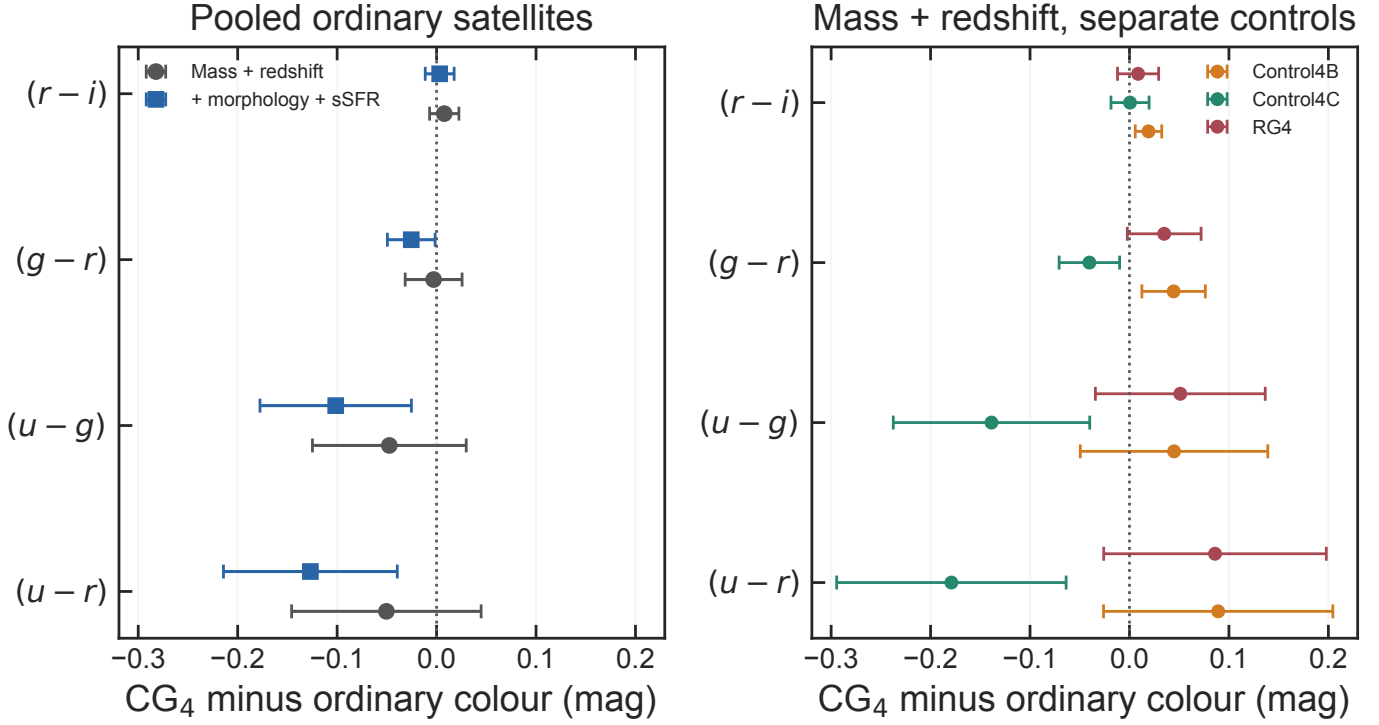


Fig. 5. Robustness of the satellite colour coefficients. Left: pooled ordinary-group comparison before and after controlling for morphology and sSFR class. Right: mass- and redshift-adjusted comparisons against each control catalogue separately. Points show the CG₄ minus ordinary colour coefficient and bars show 95% confidence intervals; negative values indicate bluer CG₄ satellites.

scale quantities, and whether they are better interpreted as compact-group-specific evolution, preprocessing in dense halo cores, or catalogue-selection effects.

5.1. Mass-, rank- and environment-adjusted galaxy properties

We fit binomial models to the combined galaxy catalogues, standardizing the continuous predictors and using group-clustered standard errors. The common adjustment set includes stellar mass, redshift, satellite/BGG rank, group luminosity and velocity dispersion when sufficiently complete.

For passive status, the CG₄ odds ratio is 1.20, with 95% confidence interval [0.88, 1.63] and $p_{\text{Holm}} = 0.485$.

For passive satellite status, the CG₄ odds ratio is 1.31, with 95% confidence interval [0.95, 1.80] and $p_{\text{Holm}} = 0.390$.

For star-forming satellite status, the CG₄ odds ratio is 0.76, with 95% confidence interval [0.56, 1.05] and $p_{\text{Holm}} = 0.390$.

For elliptical morphology, the CG₄ odds ratio is 1.62, with 95% confidence interval [1.21, 2.16] and $p_{\text{Holm}} = 0.007$.

For elliptical satellite morphology, the CG₄ odds ratio is 1.84, with 95% confidence interval [1.34, 2.53] and $p_{\text{Holm}} = 0.001$.

For elliptical BGG morphology, the CG₄ odds ratio is 0.90, with 95% confidence interval [0.46, 1.74] and $p_{\text{Holm}} = 0.751$.

For spiral morphology, the CG₄ odds ratio is 0.62, with 95% confidence interval [0.46, 0.83] and $p_{\text{Holm}} = 0.007$.

For spiral satellite morphology, the CG₄ odds ratio is 0.54, with 95% confidence interval [0.39, 0.75] and $p_{\text{Holm}} = 0.001$.

Compact-group membership therefore remains independently associated with morphology after the available ad-

justments: CG₄ galaxies have a higher probability of being elliptical/smooth and a lower probability of being spiral/features. The satellite-only morphology coefficients are stronger than the corresponding BGG coefficient, indicating that the robust morphology signal is mainly carried by the satellite population rather than by a distinct BGG morphology. In contrast, the adjusted passive and star-forming satellite coefficients are weaker and do not provide evidence for an instantaneous star-formation difference that remains significant after correction.

5.2. Matched-control comparison

We matched 234 of the 248 available CG₄ galaxies one-to-one to 234 ordinary-group galaxies without replacement. The remaining 14 CG₄ galaxies were excluded because they lay outside the common propensity-score support or exceeded the adopted caliper. The matching required common propensity-score support, a 0.2-standard-deviation caliper, and exact rank strata. The propensity model uses logMstar, z numeric, rank, log group luminosity, velocity dispersion. Projected distance is retained as a phase-space and interaction diagnostic rather than matched away because it partly defines compactness and has limited overlap. The largest absolute standardized mean difference changed from 0.28 before matching to 0.14 afterwards. The remaining maximum imbalance is slightly above the common 0.1 heuristic for excellent balance, so we treat the matched comparison as a robustness check rather than as a fully balanced pseudo-experiment.

The matched CG₄-minus-control difference in passive fraction is 0.098, with 95% bootstrap interval [0.017, 0.175] and $p_{\text{Holm}} = 0.054$.

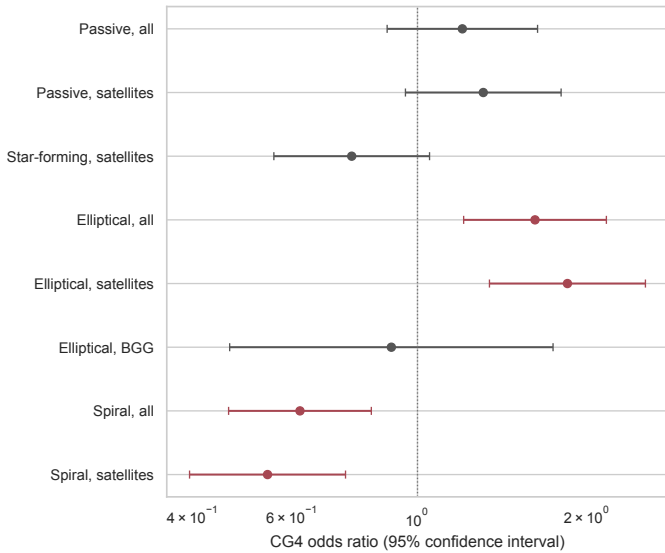


Fig. 6. Adjusted CG_4 odds ratios. Bars show 95% confidence intervals; the vertical line marks equal odds.

This is suggestive but not formally significant after correction, so we treat the passive-fraction excess as borderline.

The corresponding CG_4 -minus-control difference in star-forming fraction is -0.098 , with 95% bootstrap interval $[-0.175, -0.017]$ and $p_{\text{Holm}} = 0.054$.

The matched CG_4 -minus-control difference in elliptical fraction is 0.197 , with 95% bootstrap interval $[0.090, 0.303]$ and $p_{\text{Holm}} < 10^{-6}$.

The matched CG_4 -minus-control difference in spiral fraction is -0.197 , with 95% bootstrap interval $[-0.303, -0.090]$ and $p_{\text{Holm}} < 10^{-6}$.

Among matched star-forming galaxies, the residual-sSFR difference is -0.096 , with 95% bootstrap interval $[-0.221, 0.015]$ and $p = 0.084$.

The main interpretation does not rely on the exact GMM division of the sSFR plane. Repeating the key satellite comparison with a fixed threshold, $\log_{10}(\text{sSFR}/\text{yr}^{-1}) = -11.0$, gives CG_4 and RG_4 satellite star-forming fractions of 40.7% and 60.2%, respectively, a CG_4 - RG_4 difference of -19.6 percentage points ($p = 3.6 \times 10^{-4}$). The qualitative hierarchy is unchanged: the morphology result is independent of the sSFR threshold, whereas the passive/star-forming contrast remains more sensitive to modelling choices.

The matched comparison is therefore consistent with the adjusted models: the most robust compact-group signal is morphological, while the passive and star-forming-fraction signals are more model-dependent.

5.3. Morphology robustness to projected crowding

The morphology result could in principle be affected by projected crowding: close neighbours can make spiral structure harder to identify and can complicate SDSS deblending. We therefore repeated the raw morphology comparison after excluding galaxies with a nearest projected neighbour closer than 55 arcsec, the SDSS fibre-collision scale. Against the most compact-core-like control, Control_{4c} , the remaining CG_4 galaxies still have a higher elliptical/smooth fraction (52.5% versus 40.6%, $p = 0.001$) and a lower spiral/features fraction (37.5% versus 44.7%, $p = 0.047$). This test indicates

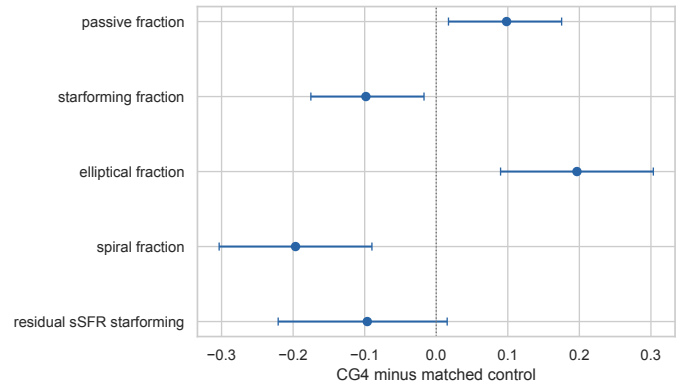


Fig. 7. CG_4 -minus-control effects after one-to-one matching. Bars show 95% bootstrap confidence intervals.

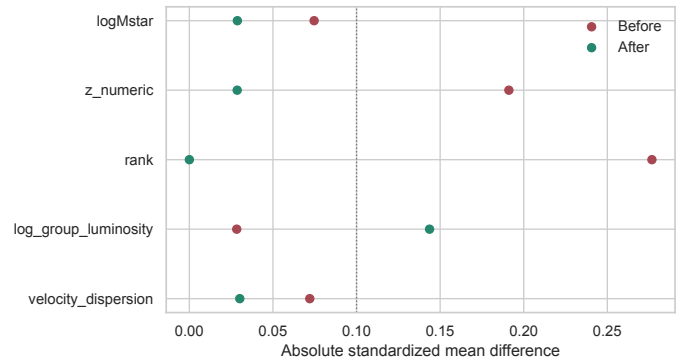


Fig. 8. Absolute standardized mean differences before and after matching.

that the raw morphology contrast is not solely produced by the very closest projected pairs.

In the adjusted models with an explicit close-neighbour flag, the CG_4 coefficient also remains significant for elliptical/smooth morphology (odds ratio 1.54, $p = 0.004$) and for spiral/features morphology (odds ratio 0.65, $p = 0.004$). The close-neighbour flag is itself strongly associated with the morphology proxy, so this test should be read as a robustness check against a simple crowding explanation rather than as a full deblending correction.

5.4. Satellite segregation around the brightest group galaxy

The previous sections show global differences between CG_4 and RG_4 satellites. We now ask whether these differences are concentrated near the BGG, as expected for accelerated evolution in the dense inner group, or whether they persist across the satellite population. The analysis uses 177 CG_4 and 166 RG_4 satellites, excludes rank-1 galaxies, ranks satellites by projected BGG-centric distance, and measures passive and early-type fractions with group-cluster bootstrap uncertainties. When available, we also use $|\Delta v_{\text{BGG}}|/\sigma_v$, with $\Delta v_{\text{BGG}} = c(z_{\text{gal}} - z_{\text{BGG}})/(1 + z_{\text{group}})$, as a projected phase-space diagnostic. Similar projected phase-space coordinates are often used as observational proxies for satellite infall state or quenching time, but only statistically (Mahajan et al. 2011; Oman & Hudson 2016; Wetzel et al. 2013).

In the innermost satellite-distance bin, the passive fractions are 68.9% for CG_4 and 48.2% for RG_4 , a CG_4 - RG_4 dif-

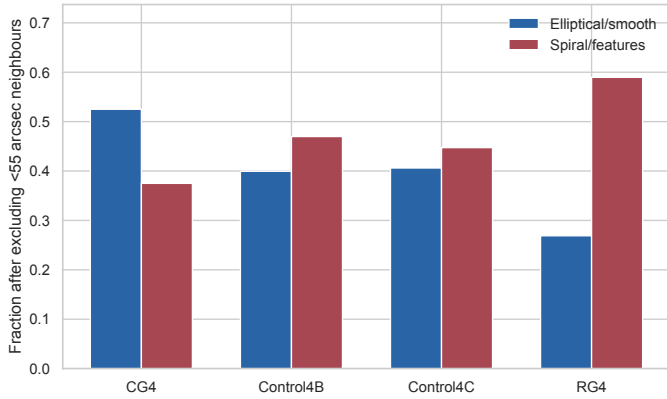


Fig. 9. Robustness of the Galaxy Zoo morphology signal to projected crowding. Galaxies with a nearest projected neighbour within the SDSS 55-arcsec fibre-collision scale are excluded in the fraction comparison, and the adjusted model includes a close-neighbour flag.

ference of 20.6 percentage points. The corresponding early-type fractions are 65.4% and 35.6%, with a difference of 29.8 percentage points.

The passive-fraction contrast is similar between the inner and outer distance ranks, so the data do not isolate the CG₄ offset to the immediate BGG environment.

After adjusting for stellar mass, redshift, distance bin, and available group-scale controls, the passive-status model gives a CG₄ odds ratio of 1.27 (95% CI 0.47–3.46; $p = 0.640$).

The analogous early-type model gives a CG₄ odds ratio of 2.57 ($p = 0.058$).

A nearest-neighbour mass–redshift matched check uses 128 CG₄–RG₄ satellite pairs. The matched passive-fraction difference is 9.4 percentage points, and the matched early-type difference is 19.6 percentage points.

Projected phase-space coordinates should not be read as orbital reconstructions for individual galaxies. In compact groups the member counts are small, velocity dispersions are noisy, and projection mixes recently accreted and long-resident satellites. We therefore treat these bins as an observational diagnostic of where the CG₄–RG₄ population contrast appears, rather than as a direct dynamical clock.

5.5. Magnitude gaps and fossil-like assembly

We define $\Delta m_{12} = M_{r,2} - M_{r,1}$ and $\Delta m_{14} = M_{r,4} - M_{r,1}$, so that a more dominant BGG has a positive, larger gap.

For Δm_{12} , the CG₄ and control medians are 1.17 and 1.09 mag, respectively ($p_{\text{Holm}} = 0.705$).

For Δm_{14} , the CG₄ and control medians are 2.56 and 2.59 mag, respectively ($p_{\text{Holm}} = 0.705$).

The compact-group morphology signal is therefore not accompanied by a significant excess in these simple fossil-like magnitude-gap indicators.

The correlation between Δm_{12} and passive satellite fraction is $\rho_S = -0.15$ ($p_{\text{Holm}} < 10^{-6}$).

The supporting magnitude-gap distributions are shown in Appendix Figs. A.19 and A.20.

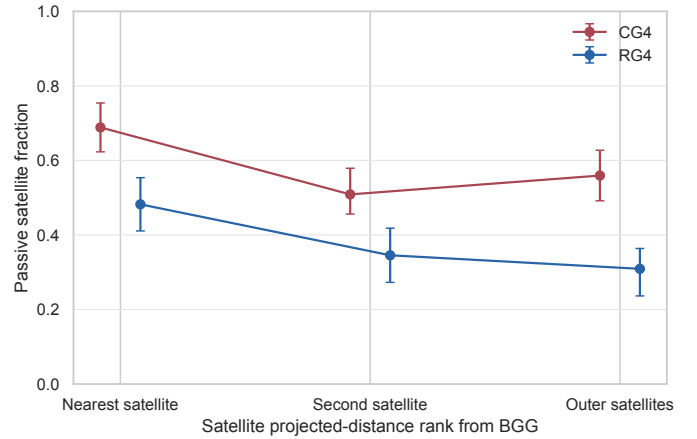


Fig. 10. Passive satellite fraction as a function of projected-distance rank from the BGG. Points show CG₄ and RG₄ fractions; bars show group-cluster bootstrap 16th–84th percentile intervals.

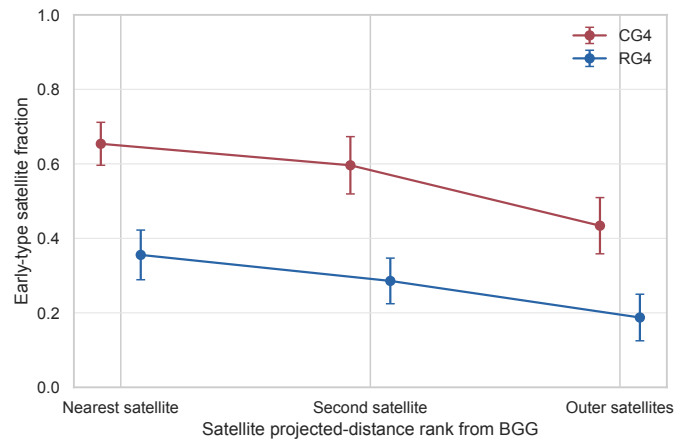


Fig. 11. Early-type satellite fraction as a function of projected-distance rank from the BGG. Because the catalogue separates ellipticals, spirals, and uncertain morphologies but not S0 galaxies, early type is equivalent to elliptical in this diagnostic.

5.6. Recently quenched candidates

The catalogues do not contain both D_n4000 and $H\delta_A$, so a post-starburst classification is not possible. We therefore perform only a limited $H\alpha$ -equivalent-width comparison, using the SDSS convention in which negative equivalent width denotes emission. We classify strong $H\alpha$ emission as $H\alpha \text{ EW} \leq -3 \text{ \AA}$.

The strong-emission fraction is 32.3% in CG₄, 41.0% in Control4B, 45.7% in Control4C, 56.9% in RG₄.

Relative to Control4B, the CG₄ strong-emission fraction differs by -0.087 ($p_{\text{Holm}} = 0.012$).

Relative to Control4C, the CG₄ strong-emission fraction differs by -0.135 ($p_{\text{Holm}} = 2.4 \times 10^{-4}$).

Relative to RG₄, the CG₄ strong-emission fraction differs by -0.246 ($p_{\text{Holm}} = 1.5 \times 10^{-6}$).

These limited $H\alpha$ results indicate that CG₄ galaxies have a lower strong-emission fraction than in each control sample in the current catalogue, but they do not provide a direct post-starburst or recently-quenched classification.

The $H\alpha$ equivalent-width distributions are shown in Appendix Fig. A.21.

5.7. Emission-line and AGN-like populations

The complementary emission-line analysis classifies BPT-star-forming, composite and broad AGN-like populations for galaxies with the required diagnostic line ratios. BPT-unclassified galaxies are excluded from the AGN-like fraction denominator rather than treated as confirmed non-AGN. Among BPT-classified CG₄ galaxies, the AGN-like fraction is 38.0%.

After mass and environment adjustment, the CG₄ AGN-like odds ratio is 1.14 ($p = 0.451$).

Thus, AGN-like activity does not provide a robust independent compact-group discriminator in this analysis.

The AGN-like fractions are shown in Appendix Fig. A.22.

5.8. Pairwise tidal index

The pairwise analysis reconstructs projected separations and velocity differences for 6268 galaxies and tests whether adding $\sum_j M_{\star,j}/R_{ij}^3$ changes the adjusted CG₄ coefficient.

For passive status, the baseline CG₄ odds ratio is 1.20 ($p = 0.227$), while the model including the tidal index gives a CG₄ odds ratio of 0.80 ($p = 0.170$). The tidal-index term itself has odds ratio 1.65 ($p < 10^{-6}$).

For elliptical morphology, the baseline CG₄ odds ratio is 1.65 ($p = 4.1 \times 10^{-4}$), whereas the model including the tidal index gives a CG₄ odds ratio of 1.15 ($p = 0.346$). The tidal-index term has odds ratio 1.56 ($p < 10^{-6}$). This suggests that the reconstructed local tidal-density proxy absorbs much of the unadjusted CG₄ morphology coefficient, although the result remains an exploratory projected-pair diagnostic.

5.9. Selection diagnostics

The fractions with complete measurements for the major derived quantities are shown in Appendix Fig. A.23. The colour-matched versus unmatched comparisons quantify shifts in stellar mass, redshift, rank, star-formation class and morphology.

The median nearest-neighbour angular separation is 100.9 arcsec in CG₄ and 209.0 arcsec in the controls; the fractions below the 55-arcsec SDSS fibre-collision scale are 19.4% and 9.3%, respectively.

No major availability difference meets both criteria of corrected significance and a difference exceeding 10 percentage points.

The group-scale availability row is based on the group covariates available for this catalogue: radius scale, velocity dispersion, group luminosity, and BGG luminosity dominance. A harmonized group-mass estimate is not available in these tables, so models that request it omit that covariate.

The colour-selection mass audit is shown in Appendix Fig. A.24.

6. Discussion: drivers of structural evolution in compact groups

6.1. Local projected tidal density and morphology signal

The most robust residual difference between CG₄ and the regular-group controls is morphological and is carried

mainly by the satellite population. CG₄ satellites are more likely to be classified as elliptical/smooth and less likely to be classified as spiral/features, while the BGG morphology coefficient is not significant after the same adjustments. We establish this excess not only relative to RG₄, but also against regular-group core analogues, especially Control_{4C}. The result hierarchy therefore points to accumulated structural transformation of the satellite population more strongly than to a present-day star-formation offset.

The pairwise tidal-index experiment strengthens this interpretation while also narrowing it. For elliptical/smooth morphology, the baseline CG₄ odds ratio is 1.65 ($p = 4.1 \times 10^{-4}$). After adding the projected pairwise tidal-index proxy, the CG₄ odds ratio decreases to 1.15 ($p = 0.346$), while the tidal-index term itself has odds ratio 1.56 ($p < 10^{-6}$). This behaviour is illustrated in Fig. 12 and is expected if compact groups are selecting galaxies in locally dense projected configurations where repeated close companions, rather than the catalogue label alone, are linked to the smooth/early-type-like population. Because the proxy is reconstructed from projected separations and catalogue masses, it should be read as a local projected-density diagnostic, not as a direct measurement of the full time-dependent tidal field. Line-of-sight alignments can inflate any R_{ij}^{-3} -weighted projected quantity, especially in samples selected to be compact in projection. There is an unavoidable element of circularity in this test: compact groups are partly selected for high projected density, and any R_{ij}^{-3} -weighted quantity is therefore related to the selection itself. Its value here is not to prove an independent causal mechanism, but to translate the categorical compact-group label into a continuous immediate-neighbour density scale. The comparison with Control_{4C} is important in this respect, because that control is also built from BGG-centred close projected companions; the tidal-index result should therefore be read as evidence that the most extreme local projected configurations, rather than group membership alone, are associated with the smooth-morphology signal.

6.2. Smooth morphologies, S0 degeneracy, and transition galaxies

The Galaxy Zoo morphology variables used here separate smooth, featureless systems from galaxies with visible spiral structure or other features. They do not cleanly distinguish classical ellipticals from S0 galaxies, faded disks, or other smooth disk-dominated systems. The enhanced CG₄ elliptical/smooth fraction should therefore be interpreted as a shift toward smooth or early-type-like appearances, not as proof that compact groups specifically build classical elliptical galaxies. The projected-crowding robustness test in Sect. 5 is therefore important: close companions can make spiral structure harder to recognize in Galaxy Zoo images, but the smooth/features signal remains present after excluding the closest projected neighbours and after adding a close-neighbour flag to the adjusted morphology model.

This caveat is important when comparing with the recent S-PLUS structural studies of compact-group galaxies by Montaguth et al. (2023, 2025, 2026b,a). Those studies identify transition galaxies and trends in the R_e –Sérsic-index and mass-size planes, with especially strong signals in non-isolated compact groups and host structures. Their transition population can plausibly overlap with the

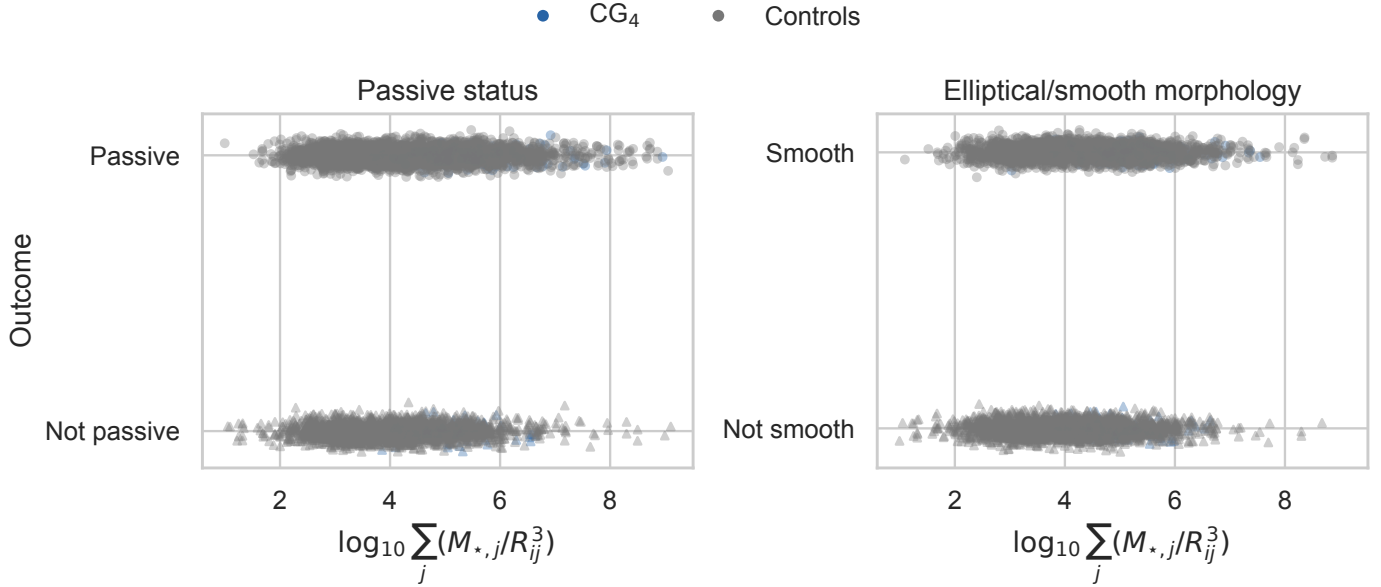


Fig. 12. Passive status and elliptical/smooth morphology as functions of the reconstructed pairwise tidal index. Colours distinguish CG₄ from control galaxies.

Diagnostic	Baseline CG ₄ signal	Survives adjusted/ matched analysis?	Interpretation
Elliptical/smooth satellite fraction	Enhanced in CG ₄	Yes	Robust accumulated satellite morphology signal.
Spiral/features satellite fraction	Reduced in CG ₄	Yes	Reduced late-type satellite vote-fraction population.
Satellite star-forming fraction	Reduced, especially versus RG ₄	Weak or borderline	Partly explained by mass, rank, environment, or preprocessing.
Star-forming main-sequence residual	Weak and control-dependent	No robust offset	No strong instantaneous SFR enhancement or suppression among remaining star-forming galaxies.
Satellite colours	Colour-slope and residual hints	Control-sensitive	Exploratory; not a simple uniform reddening or blueing of CG ₄ satellites.
AGN-like fraction	No robust independent signal	No	Not an AGN-driven explanation.
Magnitude gaps	No significant excess	No	Not fossil-like under the simple gap diagnostics used here.
Pairwise tidal index	Local projected-density proxy is informative	Absorbs much of the morphology coefficient	Morphology tracks local projected density more than the CG ₄ label alone.
Galaxy Zoo crowding check	Close-neighbour systems are more common in CG ₄	Signal persists after close-neighbour exclusion	The smooth/features excess is not solely a close-pair crowding artefact.

Table 6. Result hierarchy. Baseline tests describe where CG₄ differs from the controls; the main interpretation is based on diagnostics that remain after adjusted or matched comparisons.

smooth class in our catalogue: systems with faded disks, increasing central concentration, or S0-like structure may be counted as smooth here even if they are not classical ellipticals. The two approaches are therefore complementary. Their structural measurements describe the likely physical form of the transformation, while our regular-group controls show that the strongest residual compact-group signal is the accumulated smooth/featured morphology balance. A concentration-index check based on $R_{90,r}/R_{50,r}$ would re-

quire extending the current processed data products with homogeneous Petrosian-radius measurements and applying the corresponding quality cuts; we leave this independent structural proxy for follow-up work.

The colour diagnostics are consistent with this picture only at a suggestive level. At fixed morphology and sSFR class, the tentative bluer CG₄ satellite coefficients could indicate recently transformed or transitioning systems rather than uniformly old classical ellipticals. Because the colour-

matched subset is biased and the result is control-sample dependent, we treat this as a secondary consistency check rather than as independent evidence.

6.3. Decoupling between morphology and instantaneous star formation

The star-formation diagnostics do not follow the morphology signal one-to-one. CG₄ satellites have a lower raw star-forming fraction, especially relative to RG₄, but the adjusted and matched evidence is weaker than for morphology. Among galaxies that remain star-forming, the main-sequence residuals do not show a robust compact-group offset. The colour analysis is also control-sensitive and based on a biased colour-matched subset, while AGN-like fractions, magnitude gaps, and the limited H α diagnostic do not identify a simple alternative driver.

Spectroscopic incompleteness may also act in the same direction. The fraction of galaxies with a nearest neighbour below the 55-arcsec SDSS fibre-collision scale is 19.4% in CG₄, compared with 9.3% in the controls. This affects spectral quantities most directly, and may therefore reduce our sensitivity to the most extreme instantaneous sSFR, emission-line, or recently-quenched signatures in the closest compact-group pairs. In addition, the MPA-JHU aperture corrections rely on photometric information outside the fibre and may be less secure for interacting or blended compact-group members, so the sSFR-based diagnostics should be read with this systematic in mind. It does not remove the morphology result, which is based on imaging vote fractions, but it reinforces the need to treat the weaker star-formation conclusions conservatively.

This decoupling suggests that the dominant observable difference is not an instantaneous quenching event caught at the present epoch. A galaxy can lose visible spiral structure, build a smoother disk or bulge-dominated profile, or fade into an S0-like state on timescales that do not require a simultaneously large sSFR residual in the surviving star-forming subset. The present data therefore favour a cumulative morphology or structure signal over a single ongoing star-formation process.

6.4. Compact groups as dense cores and preprocessing sites

The comparison samples were designed to ask whether compact groups remain special relative to ordinary group cores, not merely relative to the field. The answer is mixed but physically useful. CG₄ galaxies retain a robust morphology difference, yet much of the interpretation points toward local density, dense cores, and preprocessing rather than toward an isolated compact-group channel. This is consistent with the construction result that many compact groups are embedded in or associated with larger structures, and with the stronger structural signatures reported for non-isolated compact groups in recent S-PLUS work.

The internal CG₄ class split is consistent with this interpretation but remains exploratory because the isolated subsample is small. The isolated systems contain only 6 groups and 24 galaxies, with passive and elliptical/smooth fractions of 41.7% and 29.2%, respectively. The corresponding values are 37 groups and 148 galaxies, $f_{\text{passive}} = 64.9\%$, $f_{\text{smooth}} = 51.4\%$ for embedded CG₄s, and 19 groups and 76 galaxies, $f_{\text{passive}} = 76.3\%$, $f_{\text{smooth}} = 53.9\%$ for predomi-

nant CG₄s. Combining the two non-isolated classes gives a higher passive fraction than the isolated class (68.8% versus 41.7%, $p = 0.012$). The analogous smooth-fraction contrast is suggestive rather than decisive (52.2% versus 29.2%, $p = 0.051$). We therefore use this split only as supporting consistency with a preprocessing interpretation, not as an independent proof.

The most conservative interpretation is that compact groups identify the high-density, high-interaction tail of group environments. Some systems may be genuinely compact and dynamically evolved, while others may be dense cores or substructures inside larger haloes. In both cases, the relevant physical quantity is likely the accumulated local interaction history. Future simulations and direct structural measurements, including R_e , Sérsic index, transition-galaxy classifications, and mass-size residuals, are needed to test whether the signal isolated here is unique to compact configurations or is the high-density limit of more general group preprocessing.

7. Conclusion

We have tested whether galaxies in CG₄ compact groups remain special after removing split systems and comparing them with three regular-group analogues: RG₄, Control_{4B}, and Control_{4C}. The comparison was designed to separate compact configurations from ordinary four-member groups, luminous group cores, and projected compact cores around BGGs.

The main result is morphological and is driven primarily by the satellite population. CG₄ satellites have an enhanced elliptical/smooth Galaxy Zoo fraction and a reduced spiral/features fraction, whereas the BGG morphology coefficient is not significant after adjustment. We establish that this satellite smooth/early-type excess survives not only against RG₄, but also against regular-group core analogues, including the compact-core-like Control_{4C} comparison. This signal survives the mass-, redshift-, rank-, and group-adjusted binomial models and is also recovered in the one-to-one matched-control analysis.

The star-formation result is weaker. CG₄ satellites have a lower raw star-forming fraction, especially relative to RG₄, but the adjusted and matched evidence is weaker or borderline. Among galaxies that remain star-forming, CG₄ galaxies do not show a robust offset from the fitted star-forming main sequence.

The secondary diagnostics do not identify a simple alternative driver. The colour results are control-sensitive and based on a biased colour-matched subset. AGN-like fractions and magnitude gaps do not explain the morphology signal, the H α diagnostic is limited by missing age-sensitive spectral indices, and projected phase-space trends do not localize the passive difference uniquely to the nearest satellites.

The positive result of this work is therefore not that compact groups define a separate instantaneous quenching channel, but that their satellite populations retain a smooth/early-type excess relative to regular-group core analogues, consistent with accumulated transformation in the high-density tail of group environments. The pairwise tidal-index experiment suggests that local projected tidal density accounts for part of the compact-group morphology coefficient, so the result should not be read as evidence for a uniquely isolated compact-group quenching mechanism.

Future work should connect this regular-group control framework directly to structural measurements, including R_e , Sérsic index, transition-galaxy classifications, mass-size residuals, and better spectroscopic post-starburst diagnostics. Those measurements would test whether the morphology signal isolated here is the same cumulative structural transformation highlighted by recent S-PLUS studies.

Data availability

The SDSS photometric, spectroscopic, Galaxy Zoo, stellar-mass, and star-formation quantities used in this work are publicly available through the SDSS archive. The group catalogues used for the sample construction are those of Tempel et al. (2017), Lim et al. (2017), and Díaz-Giménez et al. (2018), as described in Sect. 1 and Sect. 2. Derived catalogues and matching products will be made available through a public repository before publication and prepared for CDS/VizieR submission where appropriate.

Code availability

The analysis was performed with custom Python scripts for catalogue matching, sSFR classification, adjusted and matched comparisons, and figure generation. The code will be archived in a public GitHub or Zenodo repository before publication.

Appendix A: Supplementary diagnostic figures

This appendix collects the secondary diagnostic figures referenced in the main text.

A.1. *sSFR* and main-sequence diagnostics

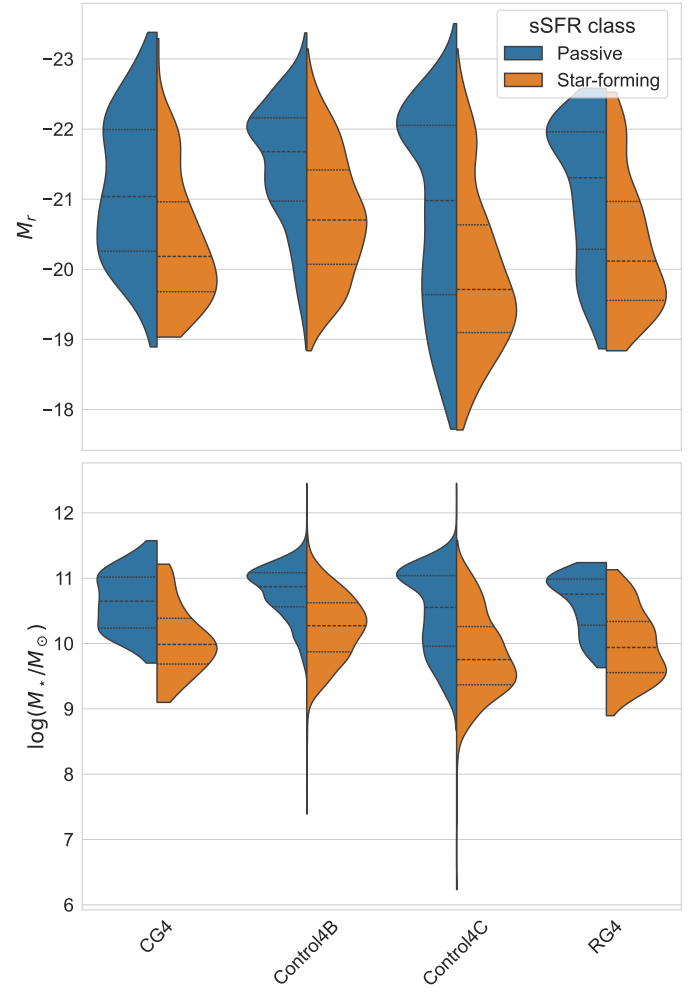


Fig. A.2. Distributions of various quantities for galaxies according to their sSFR class.

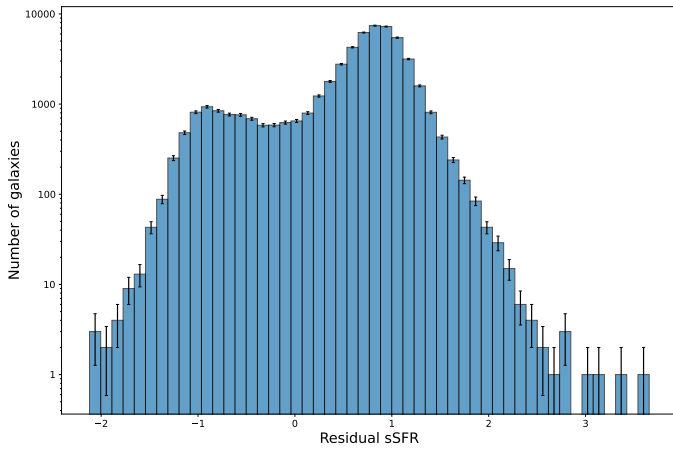


Fig. A.1. Distribution of residual sSFR offsets from the adopted star-forming main sequence.

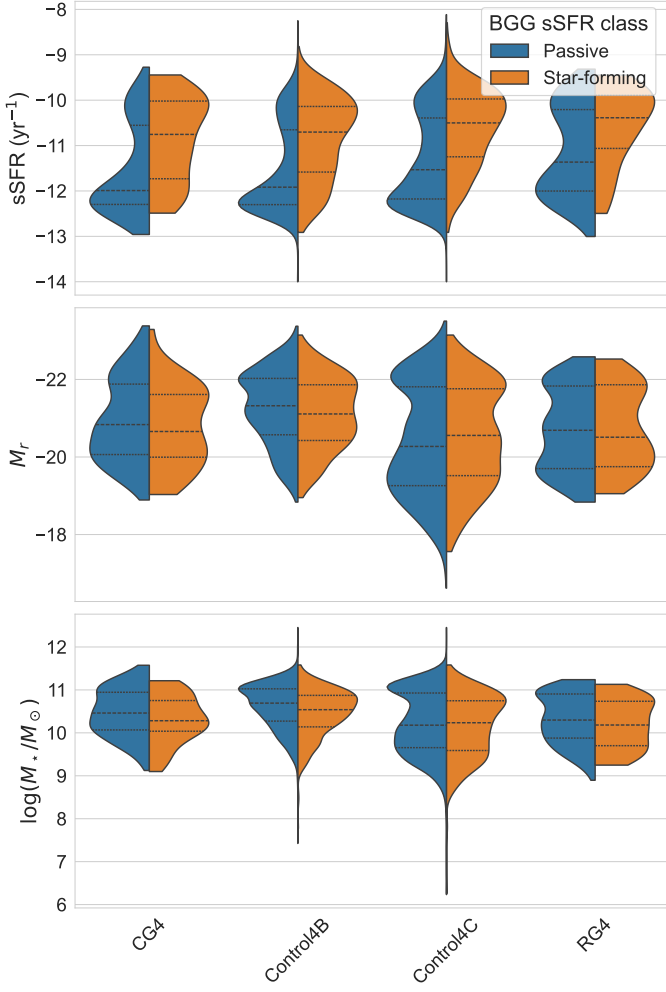


Fig. A.3. Distributions of various quantities for galaxies according to the BGG sSFR class, with BGGs included.

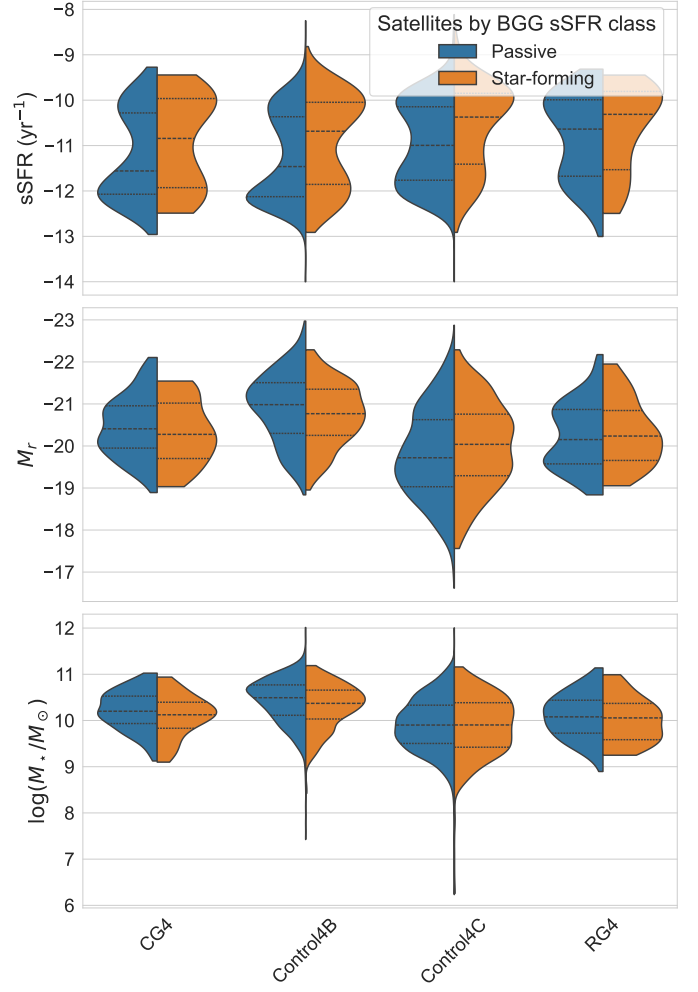


Fig. A.4. Distributions of various quantities for satellites according to the BGG sSFR class.

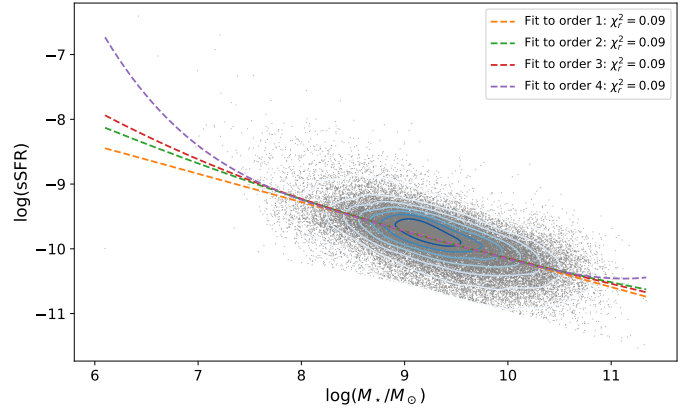


Fig. A.5. Polynomial fits of orders 1–4 to the star-forming main sequence.

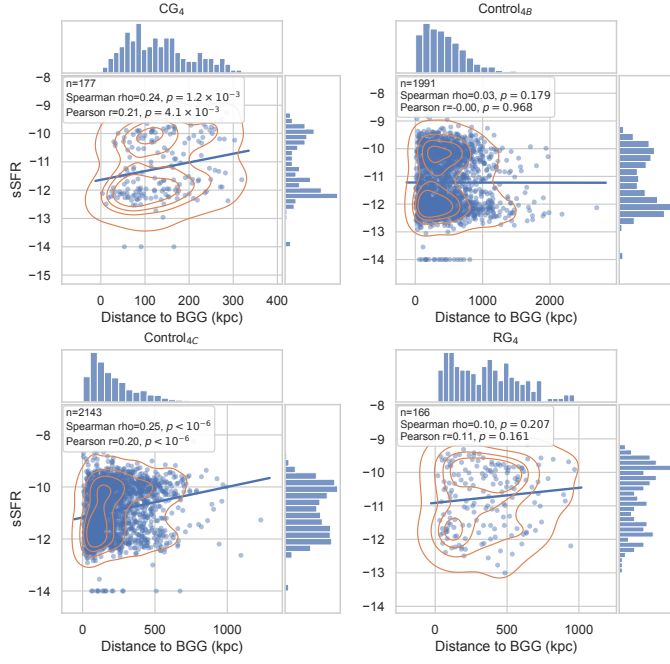


Fig. A.6. sSFR as a function of projected distance to the BGG, in kpc, for satellite galaxies in each sample.

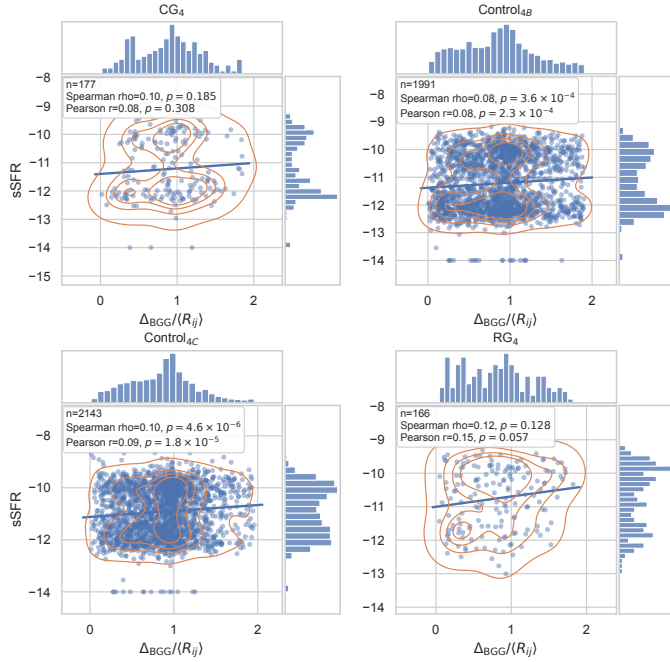


Fig. A.7. sSFR as a function of projected distance to the BGG normalized by $\langle R_{ij} \rangle$ for satellite galaxies in each sample.

A.2. Group-scale and radial diagnostics

A.3. Colour diagnostics

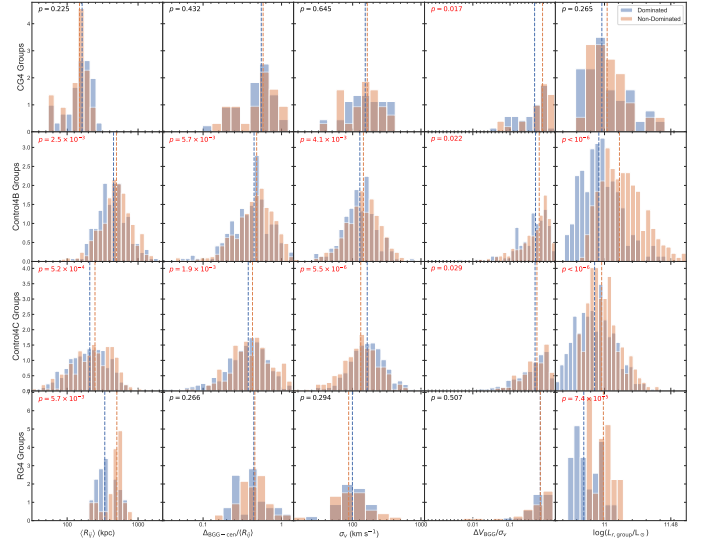


Fig. A.8. Distributions of the main group-scale observables in dominated and non-dominated systems.

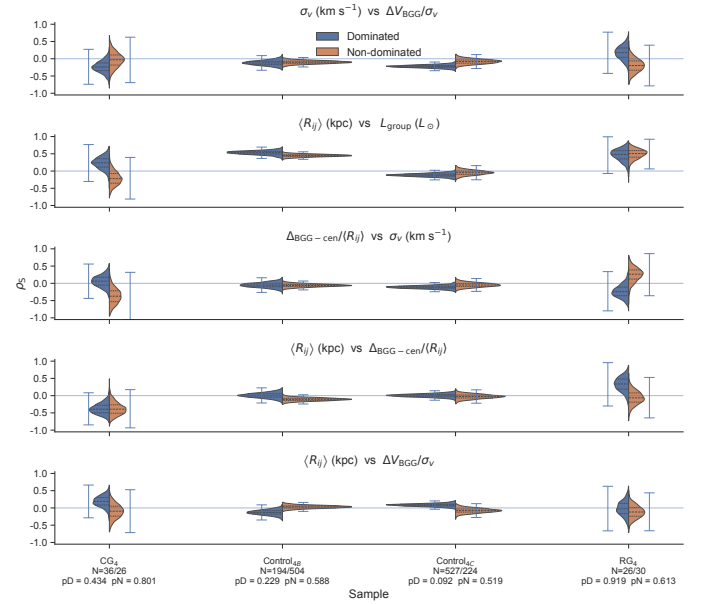


Fig. A.9. Bootstrap distributions of the Spearman coefficients for the group-property pairs selected by the XOR domination criterion.

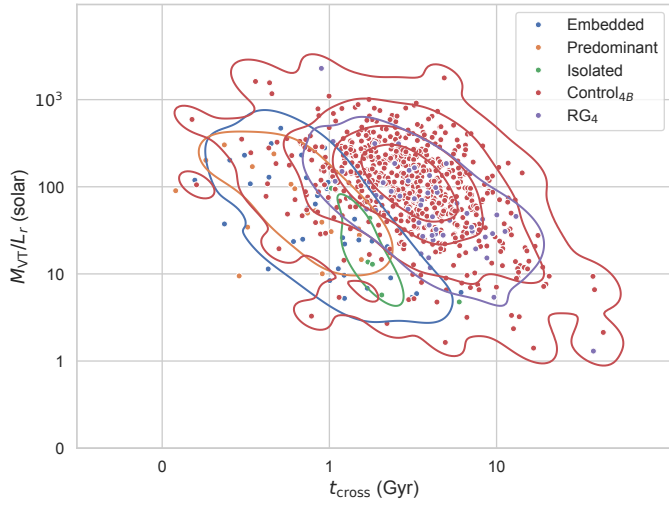


Fig. A.10. Crossing time versus virial mass-to-light ratio for the compact groups and their controls.

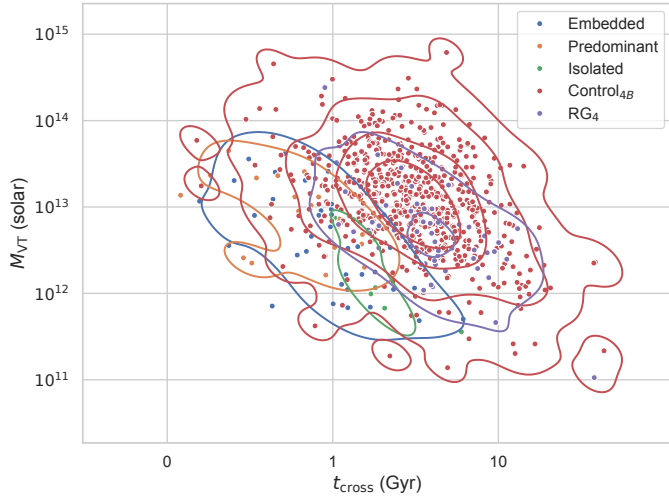


Fig. A.11. Crossing time versus virial mass for the compact groups and their controls.

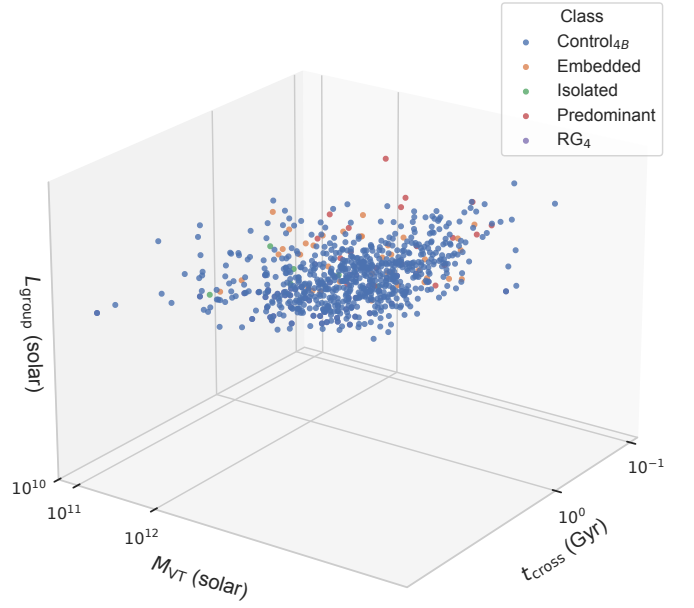


Fig. A.12. Three-dimensional view of the crossing time, virial mass, and group luminosity space.

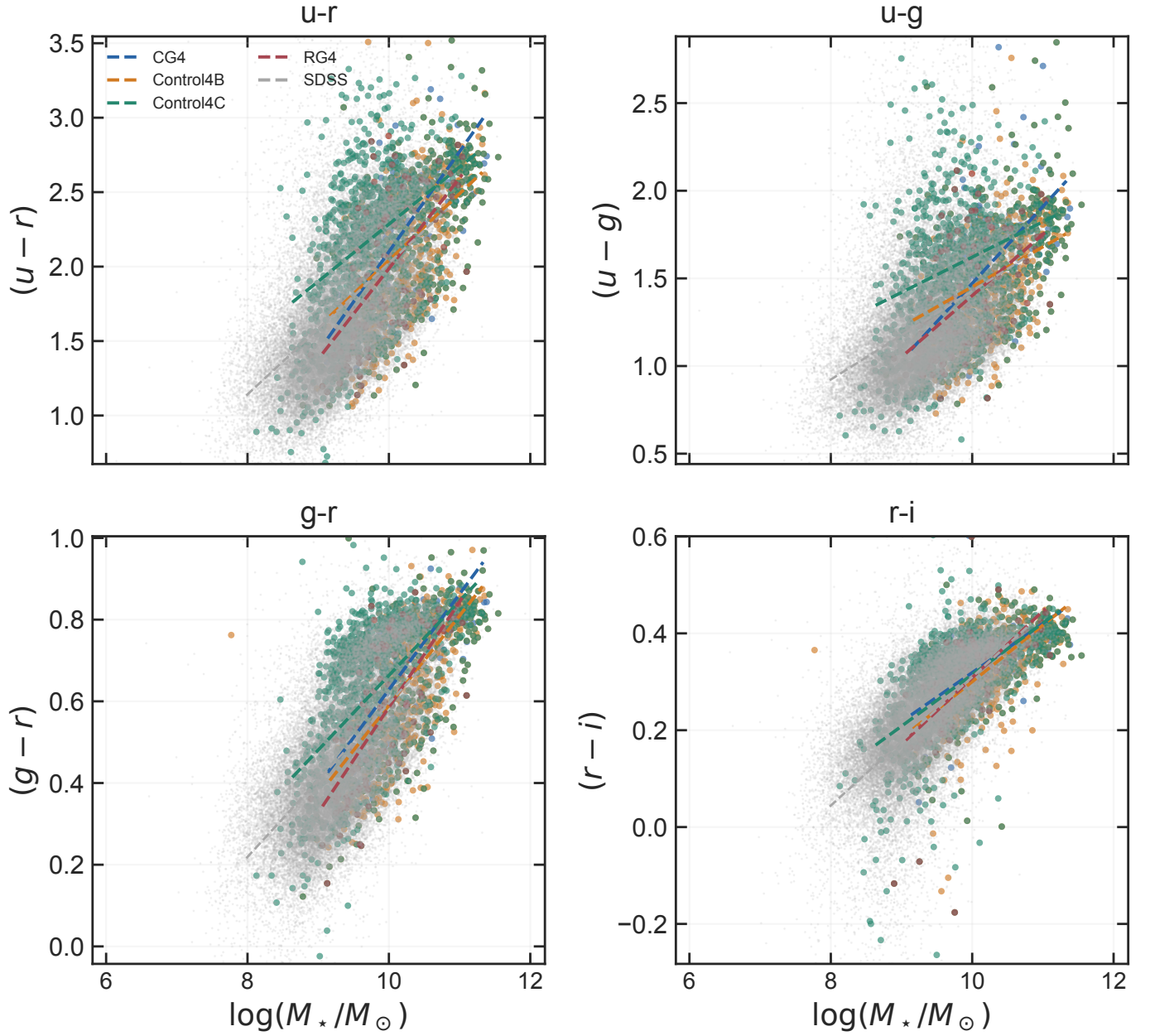


Fig. A.13. Optical colour as a function of stellar mass for CG₄, the three control catalogues, and the matched SDSS reference sample. Dashed lines show the fitted catalogue relations.

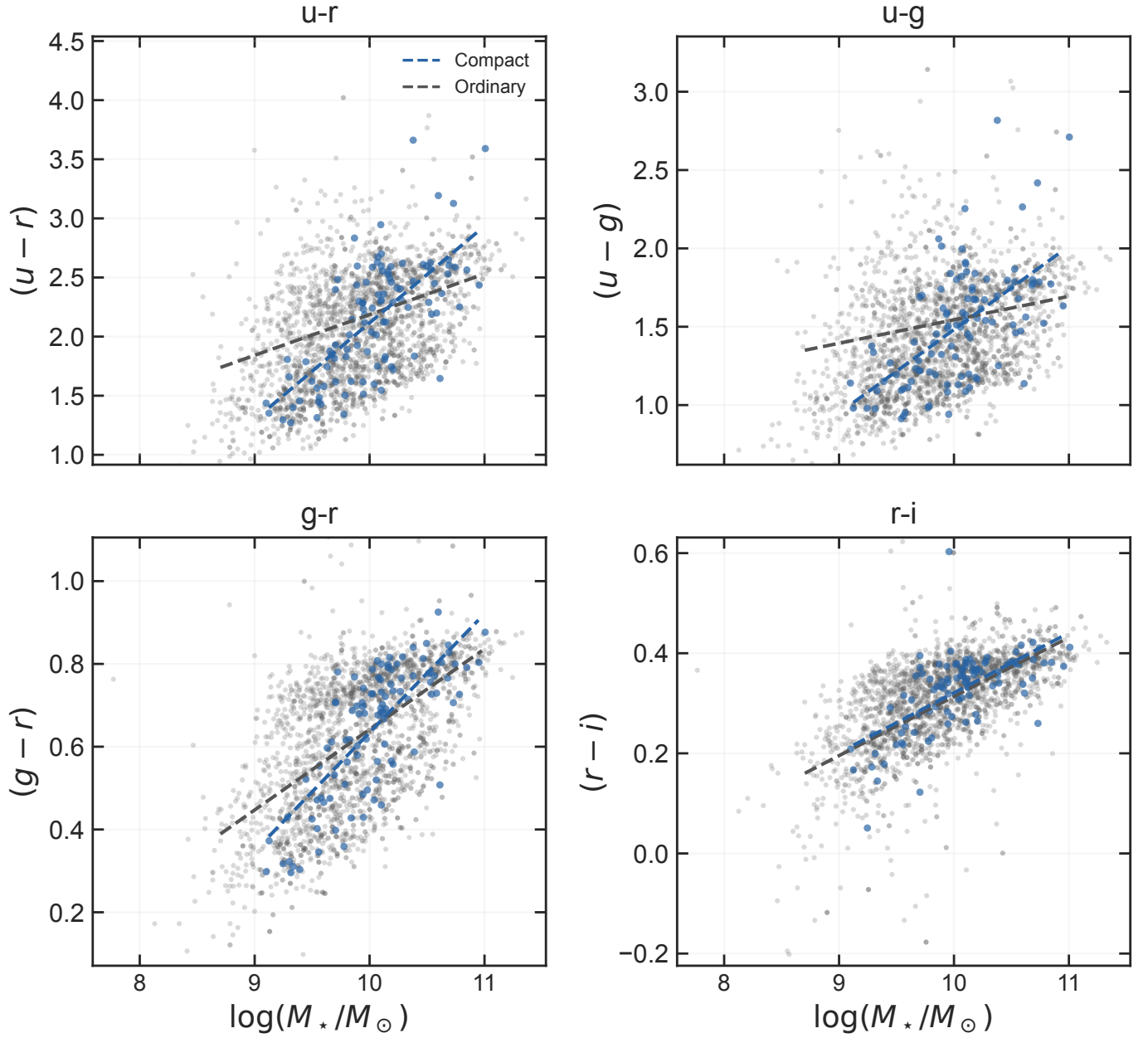


Fig. A.14. Satellite colour–mass relations in compact and ordinary groups. The fitted lines show both colour offsets at the reference mass and differences in slope.

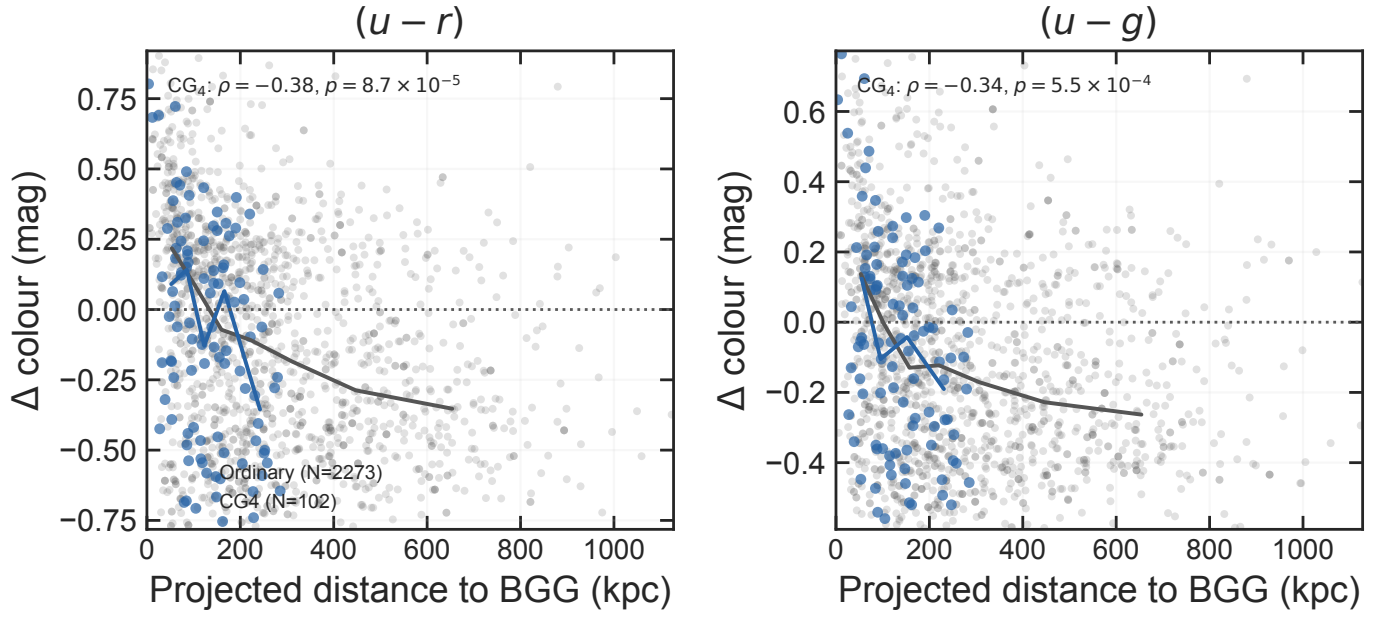


Fig. A.15. Mass- and redshift-adjusted satellite $(u-r)$ and $(u-g)$ residuals as a function of projected distance to the BGG. Lines connect binned medians for CG₄ and the pooled ordinary-group satellites; extreme plotting tails are omitted for readability, while the reported rank correlations use the full matched sample. More negative residuals are bluer than the ordinary-satellite reference relation.

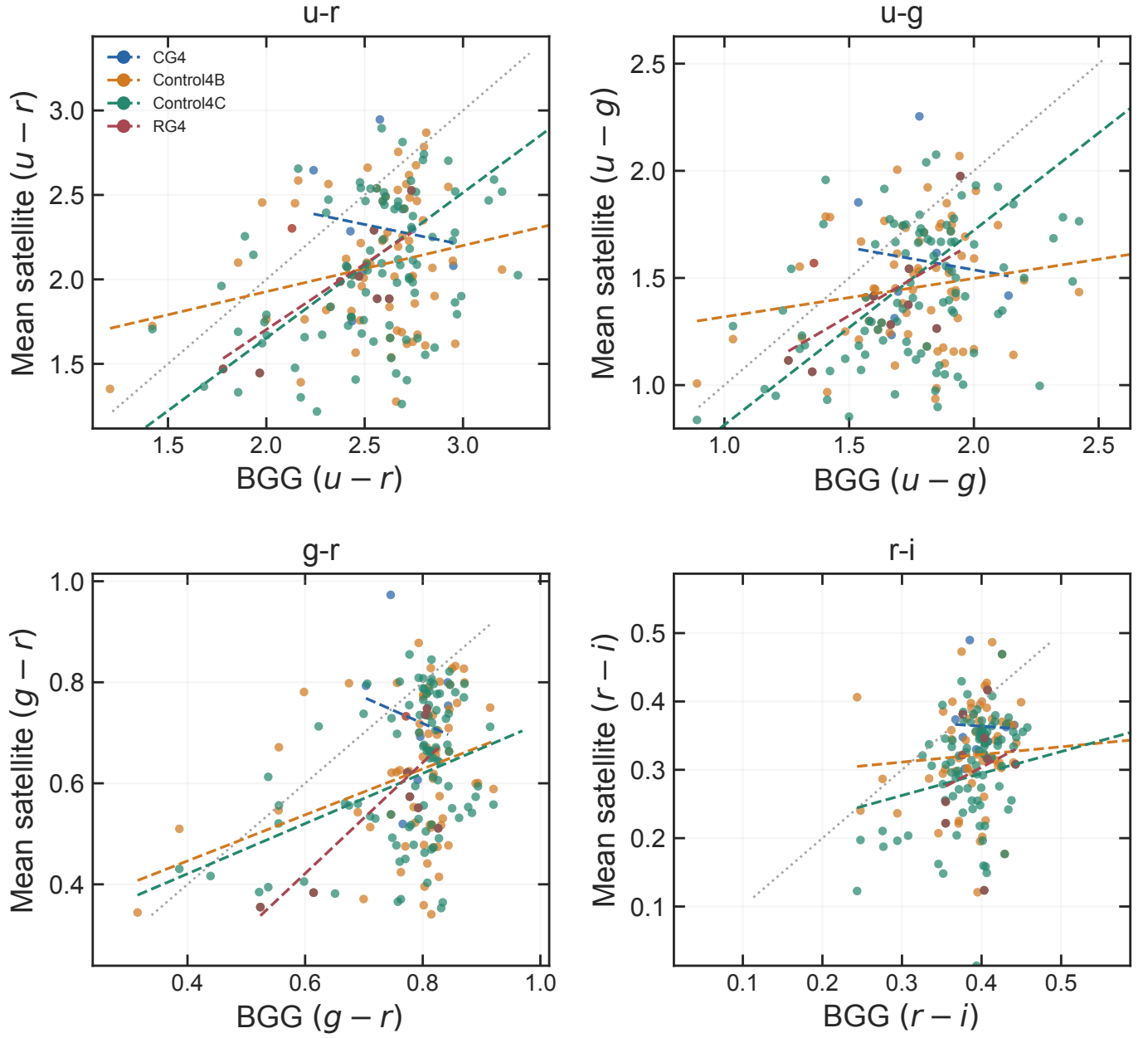


Fig. A.16. Mean satellite colour versus BGG colour for groups with matched photometry. The dotted line is the one-to-one relation and dashed lines are catalogue-specific fits.

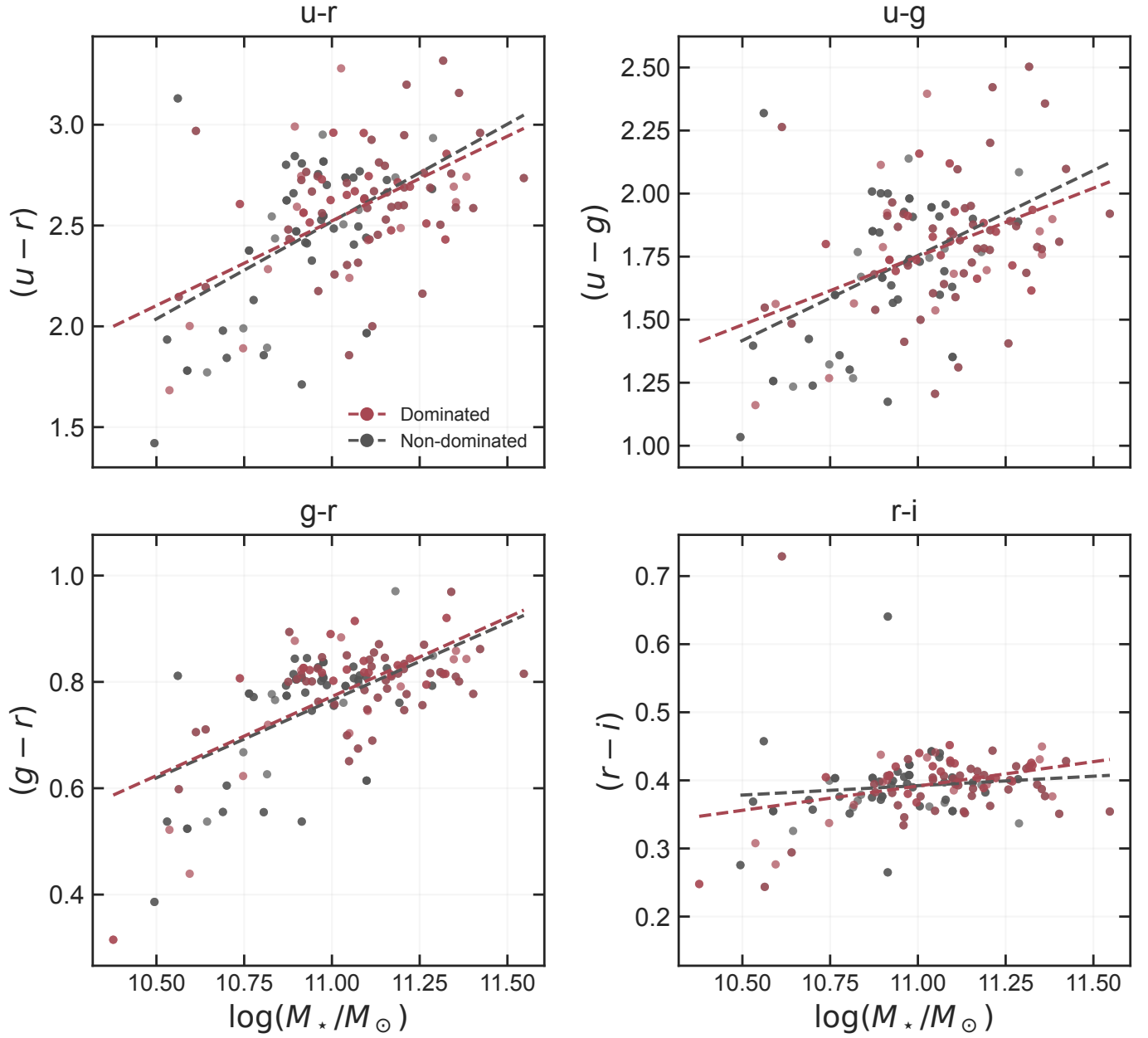


Fig. A.17. BGG colour–mass relations split by luminosity domination. Comparing the fitted relations at a common stellar mass separates colour effects from the different mass distributions.

A.4. Robustness and selection diagnostics

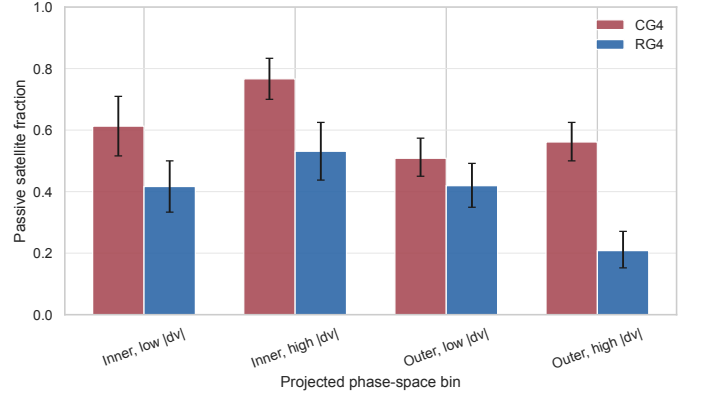


Fig. A.18. Passive satellite fraction in coarse projected phase-space bins. Inner and outer refer to projected-distance rank from the BGG; low and high velocity are split at $|\Delta v_{\text{BGG}}|/\sigma_v = 1$.

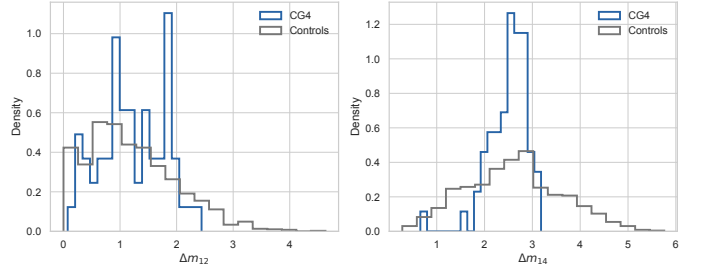


Fig. A.19. Distributions of the first-second and first-fourth magnitude gaps in CG4 and the pooled controls.

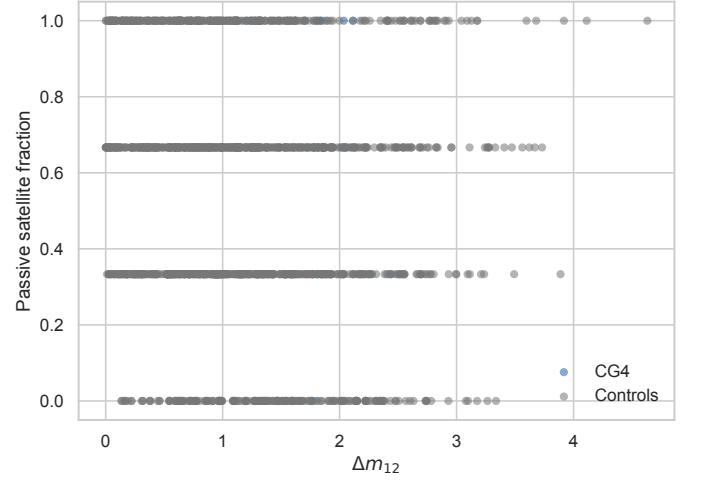


Fig. A.20. Passive satellite fraction as a function of the first-second magnitude gap.

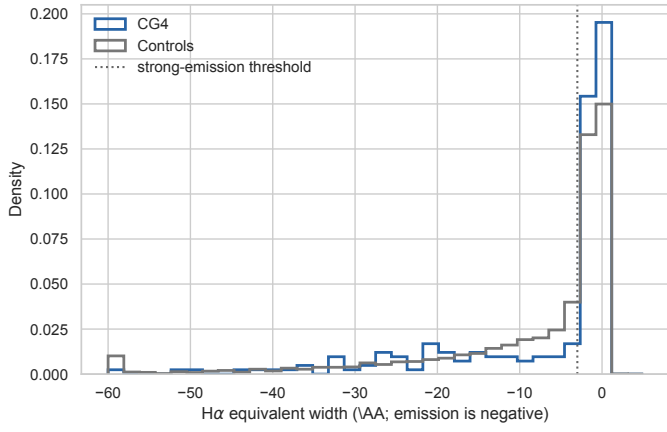


Fig. A.21. $H\alpha$ equivalent-width distributions used for the limited recent-quenching diagnostic.

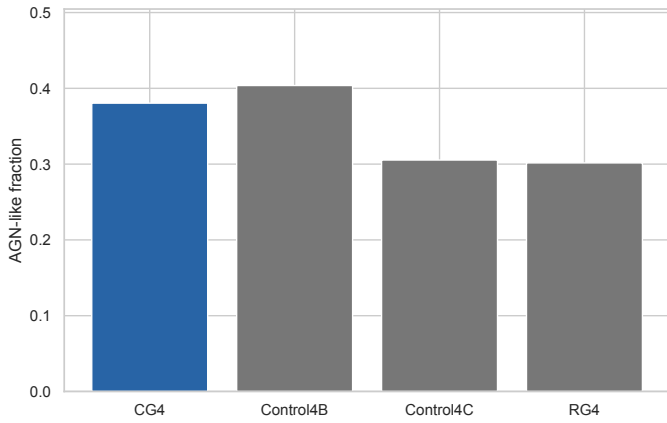


Fig. A.22. AGN-like fractions among galaxies with BPT-classifiable emission-line ratios. BPT-unclassified galaxies are not counted as non-AGN in this diagnostic.

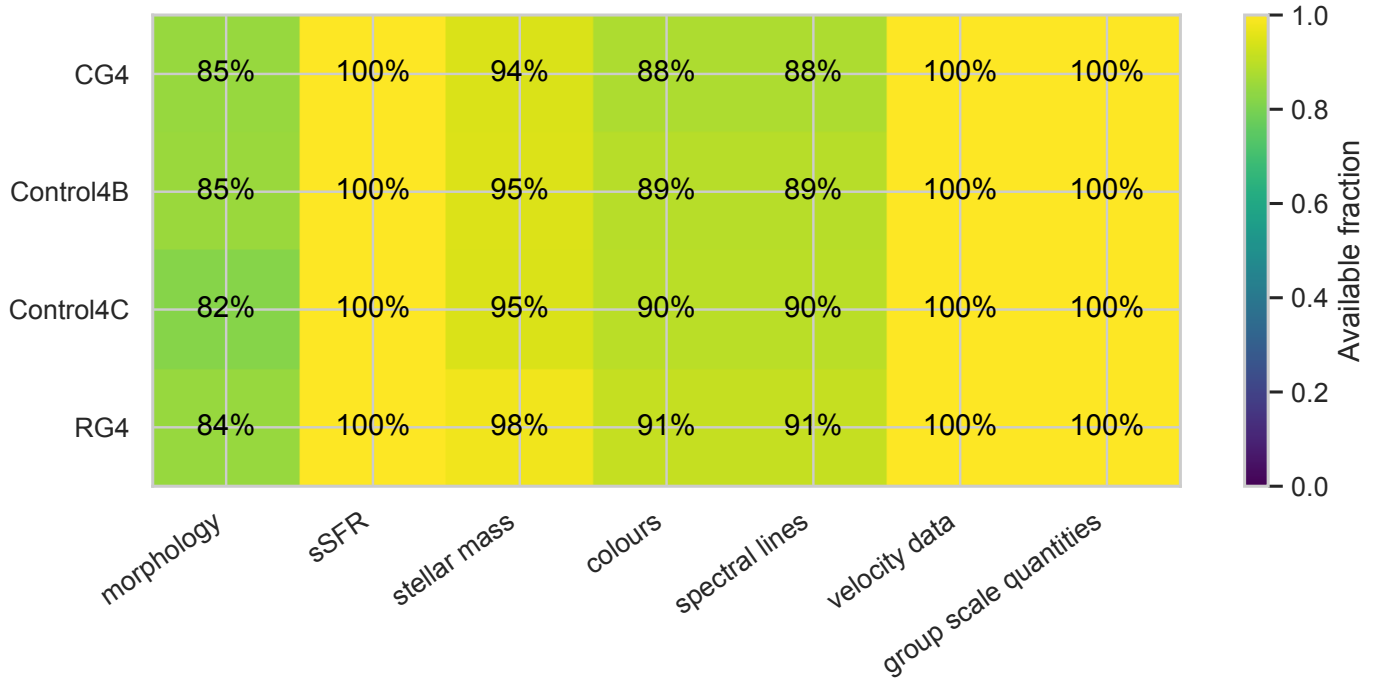


Fig. A.23. Availability fractions for the major derived quantities in each sample.

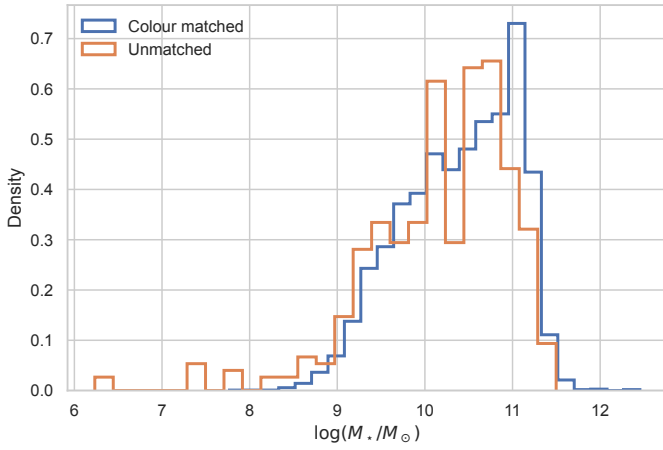


Fig. A.24. Stellar-mass distributions of galaxies with and without complete matched colour measurements.

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